END TERM EXAMINATION [MAY-JUNE 2015]
FOURTH SEMESTER [B.TECH]
POWER SYSTEM-I [ETEE-206]

Time. 3 Hours **MM : 75**

Note: *Attempt any five questions including Q.No. 1 which is compulsory. Select one question from each unit.*

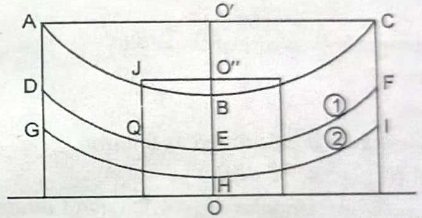
Q.1. (a) What is sag template ? Explain (5 × 5 = 25)

Ans. Normally there are two types of supports being used

(*i*) A standard or straight run tower

(*ii*) The angle or anchor tower.

In order to locate the position of the towers, the first step is to know a suitable value for the support height. The next step is to plot a sag template on a piece of transparent paper which consists of a set of curves. The horizontal and vertical distances represent the span lengths and sags respectively.



GHI (2) is the tower footing line. DEF (1) is the ground clearance line. This clearance to gound will depend upon the operating voltage and is given, according to Indian electicity rules

Q.1. (b) Explain Ferranti effect.

Ans. When a long line is operating under no load or light load condition, the receiving end voltage is greater than the sending end voltage. This is known as Ferranti effect.

Explanation:

(*i*) Assume no load condition

$$V = \frac{V_r + I_r z_c}{2} e^{\alpha x} e^{j\beta x} + \frac{V_r + I_r z_c}{2} e^{-(\alpha x + j\beta x)}$$

reduce to

$$V = \frac{V_r}{2}\left(e^{(\alpha + j\beta)} + e^{-(\alpha + j\beta)x}\right)$$

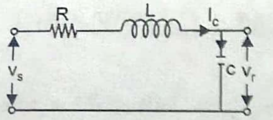
when $x = l$ and $I_r = 0$

at $l = 0$

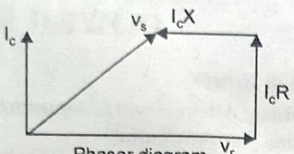
$$V = \frac{V_r}{2}(1+1) = V_r$$

As l increases, the incident component of sending end voltage increases exponentially and turns the vector anti-clock wise through an angle βL where as the reflected part of sending and voltage decreases by the same amount. The sum of these two components of sending end voltage gives a voltages which is smaller than V_r.

(ii)



Line representation under no load condition



Phasor diagram

Since usually the capacitive reactance of the line is quite large as compared to the inductive reactance, under no load or lightly loaded condition, the line load is of leading power factor. The charging current produces drop in the reactance of the line which is in phase opposition to the receiving end voltage and hence $V_s < V_r$.

Q.1. (c) Eumerate the various causes for heating of cable

Ans. The temperature rise of a body depends on the rate of generation and dissipation of heat by the body. If the rate of generation is greater than rate of dissipation, the temerature goes on rising and vice-versa. In case of an undergoing cable, the sources of heat generation are:

(i) Core loss *i.e.* copper loss in core of the cable

(ii) Dielectric loss

(iii) Sheath losses.

(i) Core loss: Resistance is calculated as follows:

(a) $$R_{es} = R_{ro}(1 + \alpha t)$$

α = temperature coefficient of conductor material.

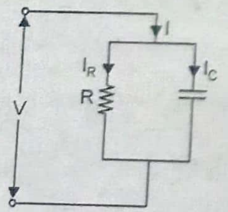
t = difference in temperature

(b) Since the effective area of section of the cable is smaller than the actual physical section, the effective resistance of the cable is larger.

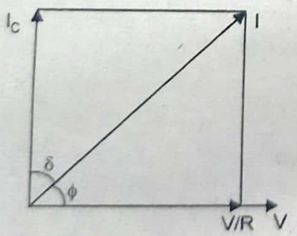
(c) The length of outermost strand is greater than the central strand and is calculated by multiplying the resistance of central strand by 1.02.

(ii) Dielectric loss:

$$P = \frac{V^2}{R}$$



$$\frac{V/R}{V_{wc}} = \tan\delta$$



$$\Rightarrow \quad \frac{V}{R} = V_{wc}\tan\delta$$

$$\Rightarrow \quad p = V \cdot Vwc\tan\delta = V^2 wc\tan\delta$$

where δ = dielctric loss angle.

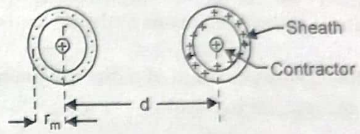
w = power supply frequency

δ <<<< (δ is very less)

$$\Rightarrow \quad \tan\delta = \delta$$

$$\Rightarrow \quad p = V^2 wc\,\delta$$

(iii) Sheath losses: When single core cable are used for a.c. transmission, the current flowing through the core of the cable gives rise to a pulsating magnetic field which when links with the sheath, incluces voltages in it. This induced voltage sets up currents under certain conditions in the sheaths and results in sheath losses.



Single Core Cable – Sheath Losses

Q.1. (d) State the advantages of bundle conductors used in a transmission system. (2)

Ans. Advantages of using bundled conductor are

(i) Reduces reactance

(ii) Reduced voltage gradient

(iii) Reduced corona loss

(iv) Reduced radio interferance

(v) Reduced surge impedance

Q.1. (e) What are the methods for improving string efficiency?

Ans. Methods of improving string efficiency are

(i) Making ratio of capacity to earth/capacity per insulator as small as possible: A long arm can help to some extent but limitation of cost and strength of towers make the ratio 1/10 as the maximum practical limit.

(ii) Grading of units: Suppose the mutual capacitance of one unit is increased progressively decreasing this figure as top unit is approached. This will result in reduction of voltage drop across the line unit since the voltage for a given current is inversely proportional to the capacitance. Capacitance a grading is very inconvenient in practice because it implies that all insulators in the string to be different from one another. This method is suitable for very high system say 220 kv.

(iii) Static Shielding: This makes use of guard ring. This ring screens the lower units and introduces a number of capacitance between the line and insulator caps. These capacitances are greater for lower units and thus voltage across those units is reduced though an equal distribution of voltages is not possible by this method.

UNIT-I

Q.2. (a) Explain Reactive power control by a synchoronous machine. (6.5)

Ans. Power transmitted from a generator the ∞ bus

$$P = \frac{|E||V|}{X}\sin\delta$$

where

E = Generator voltage

V = Infinite bus bar voltage

X = reactance of the unit

$$\delta = \text{angle between E and V.}$$

If $$E\cos\delta > |V|$$

$\Rightarrow Q > O \Rightarrow$ Generator produces reactive power. This inequality is generally satisfied when the generator is over excited and acts a capacitor. Similarly when $E\cos\delta < |V| \Rightarrow Q < 0 \Rightarrow$ Generator consumes reactive power meaning the under excited state of synchronous generator.

When machine is used over excited, it is known as synchronous capacitor and the special feature of machine is that when it runs under no load condition, there by $\delta = 0$ in above inequality.

Q.2. (b) Explain the various ratings of a circuit breaker. **(6)**

Ans. A circuit breaker is rated in terms of

(i) Number of poles

(ii) Rated voltages and current.

(iii) Rated frequency

(iv) Rated making capacity

(v) Rated symmetrical and asymmetrical breaking capacities.

(vi) Short time rating

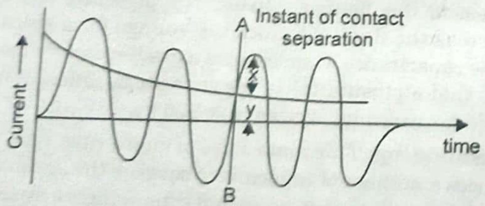
(vii) Operating duty

(i) Number of poles is a function of operating voltage.

(ii) The voltage level at various points in a power system varies depending upon system condition and as a result, the breaker has to operate under such variable voltage conditions. The breaker is expected to operate at a voltage greater than the rated nominal voltage.

(iii) The rated frequency of the breaker is the frequency for which it is designed to operate

(iv) The making current is the peak current including d.c. component in any phase during the first cycle of current when circuit breaker is closed



(v) The breaking capacity of a breaker is the product of the breaking current and the recovery voltage. The symmetrical breaking capacity is the product of symmetric breaking current and recovery voltage. Similar about asymmetric.

(vi) Short time rated current is the current that can be safely applied with the C.B. in its normal condition for 3 seconds.

Q.3. (a) Derive an expression for the sag of a transmission line supported by towers of different height at the ends. **(6)**

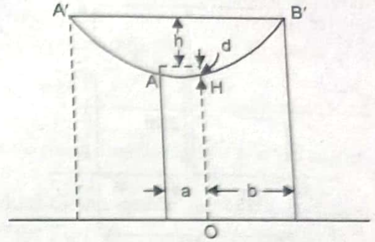
Ans.

$$y = \frac{C + x^2}{C}$$

Taking O as origin

At A $x = a \quad y = c + d$

At B $x = b \quad y = c + d + n$



$$c + d = \frac{a^2}{2c}$$

and $$c + d + h = \frac{b^2}{2c}$$

$$h = \frac{b^2 - a^2}{2c} = \frac{(b+a)(b-a)}{2c} = \frac{l(b-a)}{2c}$$

$$(b - a) = (a + b - 2a) = l - 2a$$

$$h = \frac{l(l-2a)}{2c}$$

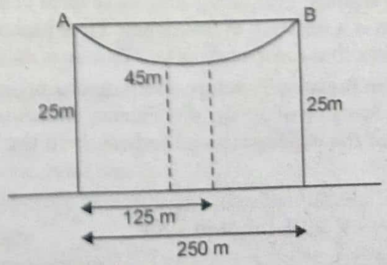
$$\frac{2ch}{l} = l - 2a$$

$\Rightarrow$ $$a = \frac{l}{2} - \frac{h}{l}$$

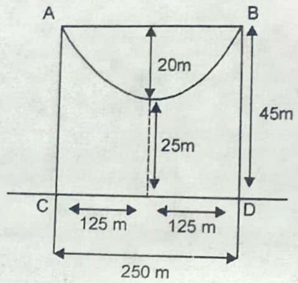
$$b = \frac{l}{2} + \frac{h}{l}$$

$$\text{sag } d = \frac{a^2}{2c} = \left(\frac{l}{2} - \frac{ch}{l}\right)^2 \cdot \frac{1}{2c}$$

Q.3. (b) An overhead transmission line at a river crossing is supported from two towers at height of 25m above the water level. The horizontal distance between the towers is 250m. If the required clearance between the conductor and water midway between the towers is 45m and if both the towers are on the same side of the point of max sag of the parabola configuration, find the stringing tension in the conductor weight of the conductor is 0.7kg/m. **(6.5)**



Ans.



$$\text{Formula of sag} = \frac{wl^2}{8T}$$

$$T = \frac{w \times l^2}{8 \times sag}$$

$$\text{Assume the Sag in '20'} = \frac{0.7 \times 250^2}{8 \times 20}$$

$$= \frac{0.7 \times 250 \times 25\cancel{0}}{8 \times 2\cancel{0}}$$

$$\boxed{T = 273.44 \text{ Kg}}$$

UNIT-II

Q.4. (a) Explain the various factors affecting corona loss. **(6)**

Ans. Corona loss formulae

$$P = 241 \times 10^{-5}\left(\frac{f+25}{\delta}\right)\sqrt{\frac{r}{d}}\,(V_p - V_o)^2\, kw/km/phase.$$

Factors affecting corona loss are

(i) Electrical factors

(ii) Atmospheric factor

(iii) Factors connected with the conductor.

(i) Electrical Factors: Frequency and wave form of supply. For corona loss, it is seen that corona loss is a function of frequency. Thus higher the frequency, higher the corona loss. This shows that corona loss is less in case of dc as compared with ac corona.

(ii) Atmospheric factors: Pressure and temperature effect. From the expression, it is clear that it is a function of air density correction factor (δ) which appears directly in the denominator of the expression and indirectly in the value of critical disruptive voltage

$$V_o = 21.1 m_0 \delta r \ln\frac{d}{r} KV$$

The lower the value of δ, the higher the loss.

Dust, Rain, snow and Hail Effect: The particles of dust clog to the conductor, thereby the critical voltage for local corona reduces which increases corona loss.

***(iii)* Factor connected with conductor** $\rightarrow$ a diameter of conductor:

$$\text{loss } \alpha \sqrt{\frac{r}{d}} \text{ and loss } \alpha\, (V - V_o)^2$$

From the expression, it is clear that the larger the dia of the conductor, the larger is corona loss

V_o is directly proportional to the size of the conductor. Hence larger the size of conductor, larger will be the corona loss.

(b) Number of conductors per phase: For operating voltages 380 kv and above, one conductor per phase gives large corona loss and hence larger radio interference level which interferes with the communication lines. This problem is solved by using bundled conductors.

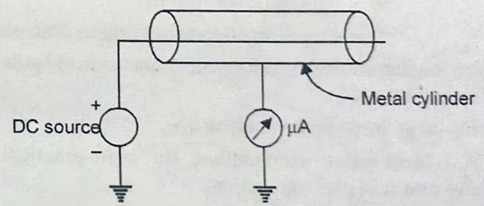
(c) Profile of the conductor: The shape of the conductor like cylinderical, oval, rectangular etc.

(d) Surface condition of the conductor: The conductors are exposed to atmospheric conditions. The dust deposited will lower the V_o and thus affects corona loss.

(e) Heating of the conductor by load current: The heating has indirect reducing effect on corona loss as heating prevents condensation on conductor surface.

Q.4. (b) Explain the phenomenon of corona. What are its effects? How can the corona loss be minimized. **(6.5)**

Ans. The ions produced by the electric field results in space charges which move around the conductor. The energy required for the charges to remain in motion is derived from the supply system. The space surrounding the conductor is lossy. In order to maintain the flow of energy over the conductor in the field where in this additional energy would have been otherwise absent.



Corona loss measurement with DC source.

$$P = 241 \times 10^{-5}\left(\frac{f+25}{\delta}\right)\sqrt{\frac{r}{d}}\,(V_p - V_o)^2 \text{ kw/km/phase}$$

(infair weather condition)

where f = supply frequency

δ = air density correction factor

r = radius of conductor, d = distance between conductors.

V_p = operating voltage

V_o = critical disruptive voltage

Method of reducing corona loss are:

(*i*) Large dia conductors

(*ii*) Hollow conductors

(*iii*) Bundled conductors

(*i*) Large dia conductors: Larger the dia of the conductors, larger will be the corona loss.

(*ii*) Hollow conductors: The idea is to have larger dia without materially adding to its weight. This conductor has a smooth surface

Disadvatages of corona

(*i*) There is a definite loss of power even though it is not much during fair weather condition.

(*ii*) When corona is present, the effective capacitance of conductors is increased because the effective dia of the conductor is increased.

Advantages of corona:

(*i*) Reduces the magnitude of high voltage steep fronted wave due to fighting or switching by partially dissipating as a corona loss.

(*ii*) Production of ozone gas

Q.5. (*a*) Explain the term surge impendence loading. What is its significance in a transmission system? (5)

Ans. Surge impedence loading is defined as the load that can be delivered by the line having no resistance, the load being at unity power factor. The power transmitted under these conditions is

$$P_r = \frac{V_r^2}{Z_0} MW$$

where V_r = Receiving end voltage in kv.

Z_0 = impedance of line in Ω

P_r = surge impedance loading or Natural power

Surge impedance loading can be used for the comparision of loads that can be carried on the lines at different voltages.

Factors affecting surge impedance loading are

(*i*) Increase in $V_r \rightarrow$ Most widely used method. But some practical difficulties related to conductor are there and it is also expensive.

(*ii*) Decrease of surge impendance $Z_0 \rightarrow$ Since the spacing between the conductors can not be decreased much, it being depedent on line voltage and corona etc the value of Z_0 can not be varied as such. But some artificial means are employed.

SIL of a power transmission line is the power delivered by a line to a purely resistive load equal to its surge impedance.

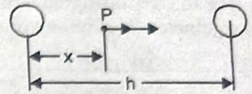
$$(I_{line}) = \frac{|V_{line}|}{3Z_0} Amp.$$

and

$$P_n = \sqrt{3}|V_L|\left(\frac{|V_L|}{\sqrt{3}Z_0}\right) = \frac{|V_2|^2}{Z_0}$$

Q.5. (*b*) Derive an expression for the capacitance of a single phase overhead transmission line. (5)

Ans.



$$V = \frac{\rho L}{\pi \varepsilon_0} \ln k$$

For this

$$C = \frac{PL}{V} = \frac{\pi \varepsilon_0}{\ln k} F/m$$

$$r = \frac{2ks}{k^2-1}$$

$$h = \frac{k^2+1}{k^2-1}s$$

$$\frac{h}{r} = \frac{k^2+1}{2k}$$

$$k^2 - 2k\frac{h}{r} + 1 = 0$$

$$k = \frac{2\frac{h}{r} \pm \sqrt{4\frac{h^2}{r^2}-4}}{2}$$

$$= \frac{h}{r} \pm \sqrt{\frac{h^2}{r^2}-1}$$

Since $\frac{h}{r} >> 1$

$$k \approx \frac{h}{r} + \frac{h}{r}$$

Since $k \neq 0$

$$c = \frac{\pi \varepsilon_0}{\frac{h}{r} + \sqrt{\frac{h^2}{r^2}-1}}$$

Q.5. (c) Explain the skin effect and proximity effect. (3.5)

Ans. When DC flows in a conductor, the current is uniformly distributed across the section of the conductor whereas flow of ac is non-uniform, with the outer filament of the conductor carrying more current than the inner. This results in higher resistance to ac than to dc and is commonly known as skin effect. This effect is more, the more is the frequency of supply and the size of conductor. The non-uniformity of flux linkage is the main cause of skin effect.

The alternating magnetic flux in a conductor caused by the current flowing in a neighbouring conductor gives rise to circulating currents which causes an apparent increase in the resistance of the conductor. This phenonmenon is called proximity effect.

UNIT-III

Q.6. A 33 kv, 3 phase, 50Hz underground line 3.4km long uses three signle core cables. Each cable has a core diameter of 2.5cm and radial thickness of 0.6cm. The relative permittivity of the dielectric is 3.1. Find. (6.5)

***(i)* Maximum stress *(ii)* Total charging KYAR**

Ans. Capacitance of cable/phase $= \dfrac{\varepsilon_r l}{41.4\log(D/d)} \times 10^{-9} F$

$$= \frac{3.1\times 3400}{41.4\times\log\left(\dfrac{3.7}{2.5}\right)}\times 10^{-9} = 1.5\mu F$$

$$\text{Voltage/phase} = \frac{33\times 10^3}{\sqrt{3}} V$$

$$\text{Charging current/phase} = \frac{Vph}{X_c} = 2\pi fC\, V_{ph}$$

$$I_C = 2\times\frac{22}{7}\times 50\times 1.5\times 10^{-6}\times\frac{33}{\sqrt{3}}\times 10^3$$

$$= 13.47 \text{ A}$$

$$\text{Charging KYAR} = 3V_{ph}\, I_C$$

$$= 3\times\frac{33}{\sqrt{3}}\times 10^3\times 13.47$$

$$= 769.91 \text{ KYAR}$$

$$g_{max} = \frac{2V}{d\log\dfrac{D}{d}}$$

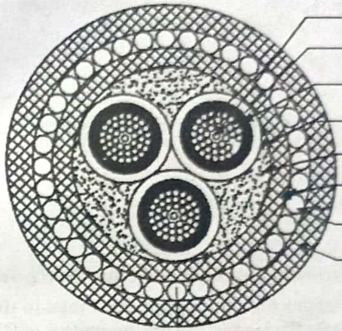
$$= \frac{2\times 33}{2.5\log\left(\dfrac{2.5+2\times 0.6}{2.5}\right)}$$

$$= \frac{2\times 33}{2.5\log\left(\dfrac{3.7}{2.5}\right)}$$

$$= 155.05 \text{ kv/cm rms}$$

Q.6. *(iii)* Explain the constructional of HSL type oil filled power cable. (6)

Ans.



This cable is the comibination of H and SL type cables. Paper insulation is provided over each core. Metalized paper is wound over insulation and lead sheath is provided over this Filler space is filled with copper woven fibre material.

Bedding armouring and serving are provided as usual.

UNIT-IV

Q.7. *(a)* Drive the expression for fault current for line to line fault (6)

Ans. Line to line fault

$$I_a = 0$$
$$I_b + I_c = 0$$
$$V_b = V_c$$

$$I_{a1} = \frac{1}{3}\left(I_a + \lambda I_b + \lambda^2 I_c\right)$$

$$I_{a2} = \frac{1}{3}\left(I_a + \lambda^2 I_b + \lambda I_c\right)$$

$$I_{a0} = \frac{1}{3}\left(I_a + I_b + I_c\right)$$

For L-L fault

$$I_{a1} = \frac{1}{3}\left(\lambda - \lambda^2\right) I_b$$

$$I_{a2} = \frac{1}{3}(\lambda^2 - \lambda) I_b$$

$$I_{a0} = 0$$

$$V_b = V_{ao} + \lambda^2 V_{a1} + \lambda V_{a2}$$

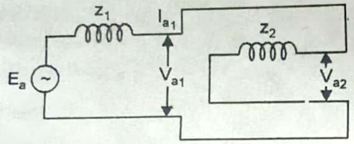
$$V_c = V_{ao} + \lambda V_{a1} + \lambda^2 V_{a2}$$

$$\Rightarrow \quad V\not{a}o + \lambda^2 V_{a1} + \lambda V_{a2} = V\not{a}o + \lambda V_{a1} + \lambda^2 V_{a2}$$

$$V_{a1} = +V_{a2}$$

$$E_a - I_{a1} z_1 = -I_{a2} z_1 = I_{a1} z_2$$

$$I_{a1} = \frac{E_1}{z_1 + z_2}$$



Q.7. (b) Explain the use of current limiting reactors in a power system (4)

Ans. Faults currents may be larger enough to cause damage to the line and other equipment of a power system network. The interrupting capacities of CB to handle such currents would also be very large.

If the system is larger or some of the generators are old, the fault current can be be kept within safe limits by increasing the system reactance. This is done by connecting reactors at strategic points in the system. Current limiting reactors are coils used to limit current during fault conditions. Such reactors have larger values of inductive reactance and low ohmic resistances.

For these, it is important that saturation at high current does not reduces the coil reactances. Air cored or iron cored reactors can be there

Types
- Air Cored
  - Dry type reactor
  - Oil immersed reactor
- Oil immersed

Q.7. (c) Enumerate the advantages of p.u. system (6)

Ans.

(1) Calculations are simplified

(2) For circuits connected by transformer, per unit system is particularly suitable. By choosing suitable base KV's for the circuits the per unit reactance remain the same, referred to either sides of the transformer.

(3) Machine reactances given in per unit, give a basis for comparison. The micro-machines are built to represent the actual machines for the purpose of research.

UNIT-IV

Q.8. (a) Give a comparison between various power flow methods. (6)

Ans. Since Gauss-siedel is undoubtedly superior to gauss method, the comparison is restricted only between GG method and NR method and that too when y bus matrix is used for problem formulation.

Mathematical background of NR method

$$y_1 = f_1(x_1, x_2, \ldots\ldots, x_n)$$

$$y_2 = f_2(x_1, x_2, \ldots\ldots, x_n)$$

$$y_{n'} = f_n(x_1, x_2, \ldots\ldots, x_n)$$

let the appx. solicition be

$$x_1^o, x_2^o, x_3^o \ldots\ldots, x_n^o$$

$$y_1 = f_1\left(x_1^o + \Delta x_1^o, x_2^o + \Delta x_2^o + \ldots.. + x_n^o + \Delta x_n^o\right)$$

$$= f_1\left(x_1^o, + x_2^o, \ldots\ldots, x_n^o\right) + \Delta x_1^o \left.\frac{\partial f_1}{\partial x_1}\right|_{x^o} + \Delta x_n^o \left.\frac{\partial f_1}{\partial x_2}\right)_{x^o}$$

$$+ \ldots. + \Delta x_n^o \left.\frac{\partial f_1}{\partial x_n}\right|_{x^o} + \phi_1$$

Applied to P.S.

$$\begin{bmatrix} \Delta P_2 \\ \Delta P_3 \\ \vdots \\ \Delta P_n \\ \Delta Q_2 \\ \Delta Q_3 \\ \vdots \\ \Delta Q_n \end{bmatrix} = \begin{bmatrix} \frac{\partial P_2}{\partial e_2} & \cdots & \frac{\partial P_2}{\partial e_n} & \vdots & \frac{\partial P_2}{\partial f_2} & \cdots & \frac{\partial P_n}{\partial f_n} \\ \frac{\partial P_3}{\partial e_2} & \cdots & \frac{\partial P_3}{\partial en} & \vdots & \vdots & & \vdots \\ \frac{\partial P_n}{\partial e_n} & \cdots & \frac{\partial P_n}{\partial e_n} & \vdots & \frac{\partial P_n}{\partial f_2} & \cdots & \frac{\partial P_n}{\partial f_n} \\ \frac{\partial Q_2}{\partial e_2} & \cdots & \frac{\partial Q_2}{\partial e_n} & \vdots & \frac{\partial Q_2}{\partial f_2} & \cdots & \frac{\partial Q_2}{\partial f_n} \\ \frac{\partial Q_3}{\partial e_2} & & \vdots & \vdots & \vdots & & \vdots \\ \frac{\partial Q_n}{\partial e_2} & \cdots & \frac{\partial Q_n}{\partial e_n} & \vdots & \frac{\partial Q_n}{\partial f_2} & \cdots & \frac{\partial Q_n}{\partial fn} \end{bmatrix} \begin{bmatrix} \Delta e_2 \\ \vdots \\ \Delta e_n \\ \Delta f_2 \\ \vdots \\ \Delta f_n \end{bmatrix}$$

$$\Rightarrow \quad \begin{bmatrix} \Delta P \\ \hline \Delta Q \end{bmatrix} = \begin{bmatrix} J_1 & J_2 \\ \hline J_3 & J_4 \end{bmatrix} \begin{bmatrix} \Delta e \\ \hline \Delta f \end{bmatrix}$$

In case of systems containing all types of buses

$$\begin{bmatrix} \Delta P \\ \hline \Delta Q \\ \hline |\Delta V_P|^2 \end{bmatrix} = \begin{bmatrix} J_1 & J_2 \\ \hline J_3 & J_4 \\ \hline J_5 & J_6 \end{bmatrix} \begin{bmatrix} \Delta e \\ \hline \Delta f \end{bmatrix}$$

J_1 off diagonal elements

$$\frac{\partial P_p}{\partial e_q} = e_p G_{pq} - f_p B_{pq} \quad q \neq p$$

diagonal

$$\frac{\partial P_p}{\partial e_p} = 2e_p G_{PP} + \sum_{\substack{q=1 \\ q \neq p}}^{n} \left(e_q G_{pq} + f_q B_{pq}\right)$$

The time taken to perform one iteration of compution is relatively smaller in case of G-S method as compared to N-R method but the number of iteration required by G.S method for a particular system are greater as compared to N-R method and they increase with the increase in size of the system.

In case of N-R system, the number of iteration is more or less independent of the size of system and vary between 3-5 iteration.

The convergence characteristics of N-R method are not affected by selection of slack bus whereas that of GS method is sometimes very seriously affected G.S method programing is easy as compared to N-R method.

N-R method may fail in respect of some ill conditional problems.

Q.8. (*b*) Explain Newton-Raphson method for load flow study **(6.5)**

Ans. N-R method

Development of load flow equations

At any bus p

$$P_p - JQ_p = V_p{}^* I_p$$

$$= V_p^* \sum_{q=1}^{n} y_{pq} V_q$$

Let

$$V_p = e_p + jf_p$$

$$y_{pq} = G_{pq} - jB_{pq}$$

$$P_p - jQ_p = (e_p + jf_p)^* \sum_{q=1}^{n} (G_{pq} - jB_{pq})(e_q + jf_q)$$

$$= (e_p - jf_p) \sum_{q=1}^{n} (G_{pq} - jB_{pq}) + f_p (e_q + jf_q)$$

$$P_p = \sum_{q=1}^{n} \{e_p (eqG_{pq} + f_q B_{pq}) + f_p (f_q Gpq - e_q B_{pq})\}$$

$$Q_p = \sum_{q=1}^{n} \{f_p (e_q G_{pq} + f_q B_{pq}) - e_p (f_q G_{pq} - e_q B_{pq})\}$$

$$(V_p)^2 = e_p{}^2 + f_p{}^2$$

J_2 elements

off diagonal

$$\frac{\partial Q_p}{\partial e_q} = e_p B_{pq} + f_p G_{pq} \quad q \neq p$$

diagonal

$$\frac{\partial Q_p}{\partial e_p} = 2e_p B_{pp} - \sum_{\substack{q=1 \\ q \neq p}}^{n} (f_q G_{pq} - e_q B_{pq})$$

J_3 elements

off diagonal

$$\frac{\partial Q_p}{\partial e_q} = e_p B_{pq} + f_p G_{pq} \quad q \neq p$$

Diagonal

$$\frac{\partial Q_p}{\partial e_q} = 2e_p B_{pp} - \sum_{\substack{q=1 \\ q \neq p}}^{n} (f_q G_{pq} - e_q B_{pq})$$

J_4 elements off diagonal

$$\frac{\partial Q_p}{\partial f_q} = -e_p G_{pq} + f_p B_{pq} \quad q \neq p$$

Diagonal elements

$$\frac{\partial Q_p}{\partial f_p} = 2f_p B_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} (e_q G_{pq} + f_q B_{pq})$$

Q.9. (*a*) Describe the classification of buses in a power system. **(6)**

Ans. Bus classification

= Load bus

= Generator bus

= Slack bus

(1) Load bus: At this bus, real and reactive components of power are specified. It is desired to find out the voltage magnitude and phase angle at such a bus as at a load bus voltage can be allowed to vary with in the permissible values eg 5%.

(2) Generator bus or voltage controlled bus: There the voltage magnitude corresponding to the generation voltage and the real power p_G corresponding to its ratings are specified. It is required to find out the reactive power generation Q_G as phase angle of bus voltage.

(3) Slack, Swing or reference bus: At this bus, the voltage magnitude v and phase angle δ are specified whereas real and reactive powers P_G and Q_G are obtained through load flow solution.

Bus type	Quantities specified	Quantities to be obtained
Load Bus	P, Q	|V|, δ
Generator bus	P, |V|	Q, δ
Slack bus	|V|, δ	p, Q

Q.9. (*b*) Explain Gauss-siedel method for load flow study. What are its advantages and diadvantages? **(6.5)**

$$V_2^{K+1} = \frac{1}{y_{22}} \left[\frac{p_2 - jQ_2}{(V_2^k)^*} - (y_2, V_1 + y_{23} V_3^k) \right]$$

$$V_3^{K+1} = \frac{1}{y_{33}}\left[\frac{P_3 - jQ_3}{(V_3k)^*} - \left(y_{31}V_1 + y_{33}V_3^K\right)\right]$$

$$Vp^{K+1} = \frac{1}{y_{pp}}\left[\frac{Pp - jQp}{(V_P^k)^*} - \sum_{q=1}^{p-1} 4_{pq}Vq^{k+1} - \sum_{q=p+1}^{n} y_{pq}V_q^k\right]$$

Points to remember

(i) voltage and admittance are complex quantities

(ii) Buses 2, 4, 5 are load buses 3 is a voltage controlled bus

$$Q_p = \sum_{q=1}^{n}\left\{f_p\left(e_qGp_q + f_qB_{pq}\right) - e_p\left(f_qG_{pq} - e_qB_{pq}\right)\right\}$$

where

$$Y_{pq} = G_{pq} - jB_{pq}$$
$$V_p = e_p + j_{fp}.$$

Advantages:

(i) Ease of programming

(ii) Accurate

(iii) Better Convergence

Disadvantages:

(i) Time consuming

(ii) Large number of iterations

FIRST TERM EXAMINATION [FEB–2016]
FOURTH SEMESTER [B.TECH]
POWER SYSTEM-I [ETEE–206]

Time : 1½ Hrs. **M.M. : 30**

Note: *Q.No.1 which is compulsory and any two more questions from remaining.*

Q.1. (a) Explain why there is an increase in capacitance due to bundling of conductors? (2)

Ans. For the same GMD, because of the bundling of conductors, GMR increases which leads to the higher capacitance of the bundled conductor. It also reduces the voltage gradient in the vicinity and hence helps in reduced corona losses.

Q.1. (b) What is surge impedance and surge impedance loading of the line? (2)

Ans. Surge impedance of a line is given by $Z_0 = \sqrt{\frac{L}{C}}$. It is the ratio of the amplitudes of voltage and current of a single wave propagating along the line. *i.e.* a wave travelling in one direction in the absence of reflection in the other direction.

Surge impedance loading (SIL) of a transmission line is the travelling MW loading of a transmission line at which a natural reactive power balance occurs.

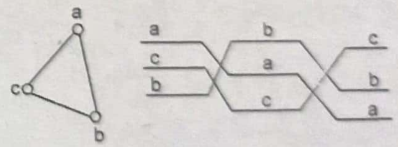
Q.1. (c) What are skin and proximity effects? (2)

Ans. Skin Effect: When DC flows in the conductor, the current is uniformly distributed across the section of the conductor whereas flow of AC is non-uniform, with the outer filaments of the conductor carrying more current than the filaments closer to the centre. This results in a higher resistance to AC than to DC and is commonly knows as Skin Effect.

The alternating magnetic flux in a conductor caused by the current flowing in a neighbouring conductor gives rise to circulating current which cause an apparent increase in resistance of a conductor. This phenomenon is called Proximity effect.

Q.1. (d) What is transposition of transmission lines? (2)

Ans. By transposition of conductors is meant the exchaning of position of the power conductors at regular intervals along the line, so that each conductor occupies the original position of every other conductor over an equal distance.



Q.1. (e) Explain clearly the Ferranti effect phasor diagram. (2)

Ans. When a long line is operating under no load or light load condition, the receiving end voltage is greater than the sending end voltage. This is known as Ferranti effect.

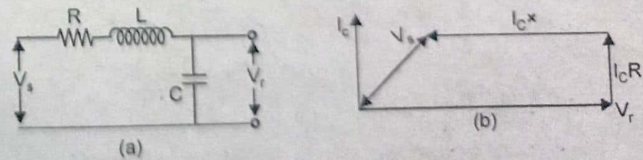


Fig. (a) Line representation under no load (b) Its phasor diagram

$$V_s = V_R + I(R\cos\phi_R + X\sin\phi_R)$$

$$6351 = V_R + I(5\times0.4 + 5.6\times0.6)$$

$$= V_R + 10.36\,I$$

$$6351 = V_R + \frac{10.36\times10^6}{2.4V_R}$$

$$V_R^2 - 6351\,V_R + 4.2\times10^6 = 0$$

$$V_R = \frac{6351\pm\sqrt{6351^2 - 4\times4.2\times10^6}}{2\times01}$$

$$= \frac{6351\pm4851.31}{2\times1}$$

$$= 5601.15 \text{ V or } 749.845 \text{ V (Not considered)}$$

$$V_R = 5601.15 \text{ V}$$

$$V_{RL} = 5601.15\times\sqrt{3} = 9701.48\text{V}$$

$$I_R = \frac{10^6}{3\times0.8\times5601.15} = 74.4\text{A}$$

$$\eta_T = \frac{\text{Output}\times100}{\text{Output + Losses}} = \frac{10^6\times100}{10^6 + 3\times74.4^2\times10/2}$$

$$= 92.33\%$$

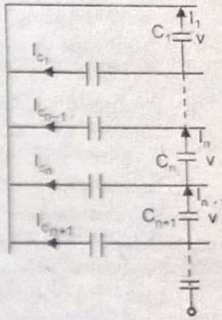
END TERM EXAMINATION [MAY–2016]
FOURTH SEMESTER [B.TECH]
POWER SYSTEM–I [ETEE–206]

Time : 3 Hrs. **M.M. : 75**

Note: *Attempt any five questions including Q.No.1 which is compulsory. Select one question from each unit.*

Q.1. (a) Explain capacitance grading for improving voltage distribution across units of string insulator. (5)

Ans. The unequal distribution of voltage is due to leakage current from the insulator pin to the tower structure. This current can't be eliminated. The other possibility is that disc of different capacitive reactance could be used s.t. product of their capacitive reactance and current flowing through respective unit is same. This requires that the unit nearest the cross arm should have minimum capacitance.



$$I_{C1} = \frac{I_{c_n}}{n}$$

$$nV\omega l = I_{cn}$$

At junctions n

$$I_{n+1} = I_n + I_{c_n}$$

$$V\omega C_{n+1} = V\omega C_n + nV\omega C$$

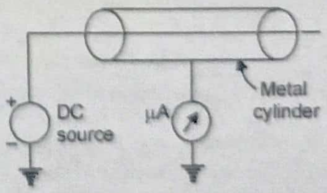
$$C_{n+1} = C_n + nC$$

If the capacitance of one unit is fixed, the capacitance of other units can be found for equal distribution of Voltage across the units of the string.

This seems that in order to carry out unit grading, units of different capacity are required. This requires large stocks of different sized units, which is uneconomical and impractical.

Q.1. (b) Define Corona Loss. (5)

Ans. The ions produced by electric field result in space charges which move round the conductor. The energy required for the charges to remain in motion is derived from the supply system. The space surrounding the system is lossy. In order to maintain the flow of energy would have been otherwise absent, it is necessary to supply this additional loss from the supply sytem. This additional power is referred to as corona loss.



Setup for corona loss measurement

$$P = 241\times10^{-5}\left(\frac{f+25}{\delta}\right)\sqrt{\frac{r}{d}}(V_p - V_0)^2 kW/km/\text{phase}$$

where P = corona loss

f = supply frequency

δ = air density correction factor

V_p = operating voltage in (kV)

V_0 = Critical disruptive voltage (kV)

r = radius of conductor

d = distance between conductors.

Factors affecting corona loss are:

(*i*) Electrical factor

(*ii*) Atmospheric factors

(*iii*) Factors connected with the conductor

(*a*) Number of conductors/phase

(*b*) Profile of the conductor

(*c*) Surface conditions of the conductor

(*d*) Heating of the conductor by load current.

Methods of reducing corona loss are:

(*i*) Large dia conductors

(*ii*) Hollow conductors

(*iii*) Bundled conductors.

Q.1. (*c*) What is the significance of using α (Alpha) operator in studying component synthesis? (5)

Ans. For convenience of notation and manipulation a phase operator (α) is introduced. Through usage it has come to be known as a phasor (α)

$$\alpha = 0.5 + j\,0.866 = e^{j2\pi/3}$$

$$\alpha^2 = (e^{j2\pi/3})^2 = e^{j4\pi/3} = e^{-j\frac{2\pi}{3}}$$

$$= -0.5 - j0.866$$

$$\alpha^3 = e^{(j^2\pi/3)^3} = e^{j2\pi} = 1 + j0$$

$$\alpha^4 = \alpha^3\,\alpha = 1.\,\alpha = \alpha$$

$$\alpha^5 = \alpha^3.\,\alpha^2 = 1.\alpha^2 = \alpha^2$$

and so on

α is numerically the cube root of unity.

Q.1. (*d*) Define per unit system and why it is used? (5)

Ans. The per unit system: In a large interconnected power system with various voltage levels and various capacity equipments it has been found quite convenient to work with per unit system of quantities for analysis purposes rather than in absolute values of quantities. The *Pu* value is defined as

$$Pu = \frac{\text{The actual value of quantity}}{\text{The base value of the quantity in same unit}}$$

$$V_{Pu} = \frac{V_{\text{actual}}}{V_{\text{base}}} \qquad Z_{pu} = \frac{Z_{\text{actual}}}{Z_{\text{base}}}$$

If in case the value has changed

$$Z_{pu\text{-}new} = Z_{pu\ old}\cdot\frac{\text{kVA}_{new}}{\text{kVA}_{old}}\left(\frac{V_{old}}{V_{new}}\right)^2 \qquad Z_{pu} = \frac{I\,Z}{V}$$

Q.1. (*e*) Explain different type of buses used in load flow analysis (5)

Ans. Various kinds of buses are

(1) Load bus

(2) Generator or voltage controlled bus

(3) Slack or swing or reference bus

Bus type	Quantities specified	Quantities to be obtained
Load bus	P, Q	|V|, δ
Generator bus	P, |V|	Q, δ
Slack bus	|V| δ	P, Q

Load bus: P,Q are defined an |V|, δ are to be obtained.

Generator bus: P, |V| are defined & Q, δ are to be obtained.

Stack bus: In a power system, there are mainly two types of buses *i.e.* load & generator buses. For these, P_i is specified. Now $\sum_{i=1}^{n} P_i$ = real power loss P_L which is taken as positive for generator bus and negative for load buses. The losses remain unknown untill the load flow solution is complete. It is for this reason that generally one of the generator buses is made to take the additional real and reactive power to supply transmission losses.

The phase angle of the voltage at slack bus is usually taken as reference.

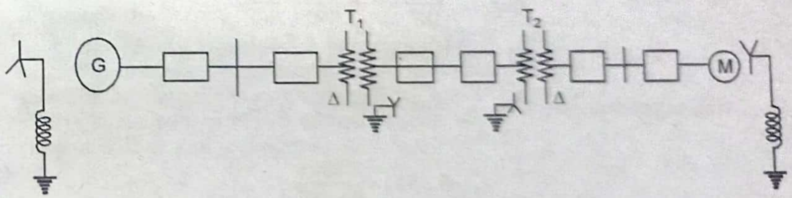
$$V_i \angle \delta_i = e_i + jf_i$$

e_i, and f_i are real and reactive component of voltage. at i_{th} bus.

UNIT-I

Q.2. (*a*) Draw and explain each component in single line diagram of power system. (6)

Ans.



□ : Circuit Breaker

Generator : 40 MVA, 11 KV, x^n = 20%

Motor : 30 MVA, 11 KV, x^n = 30%

Transformer T1 : 40 MVA, 11/220 kV, x = 15 %

Transformer T2 : 40 MVA, 220/ 11KV, x = 15%

: Single line diagram of a typical power system: The single line diagram of a power system network shows the main connections and arrangements of the system components along with their delta. In case of transmission lines sometimes the conductor size and spacing are given. It is not necessary to show all the components of the system on a SLD. In a SLD, the system components are usually drawn in the form of their symbols. Generator & transformer connections-star, delta & neutral earthing are indicated by symbols drawn by the side of the representation of these elements.

Q.2. (b) A transmission line has a span of 275 mtrs between level supports. The conductor has a diameter of 19.53 mm, weighs of 0.844 kgf/m and has an ultimate breaking strength of 7950 kgf. Each conductor has a radial covering of ice 9.53 mm thick and is subjected to a horizontal wind pressure of 40 kgf/m² of the ice covered projected area. If the factor of safety is 2, calculate the deflected sag and the vertical component of the sag. One cubic meter of ice weighs 913. 5kgf. **(6.5)**

Ans.

$$\text{Working Tension} = \frac{\text{Ultimate strength}}{\text{Factor of safety}}$$

$$= \frac{7950}{2} = 3975 \text{ kgf}$$

Weight of ice coating per meter length

$$w_i = \text{Density of ice} \times \pi r (D + r)$$

$$= 913.5 \times \frac{22}{7} \times (19.53 + 9.53) \times 9.53 \times 10^{-6}$$

$$= 0.7951 \text{ kg/m}$$

Wind force per meter length of conductor

$$w_w = P(D + 2r)$$

$$= 40\,(19.53 + 2 \times 9.53) \times 10^{-3}$$

$$= 1.5436 \text{ kg/m}$$

$$w_r = \sqrt{(w_c + w_i)^2 + w_w^{\,2}}$$

$$= \sqrt{(0.844 + 0.7915)^2 + 1.5436^2}$$

$$= 2.2515$$

$$\text{Sag} = \frac{w_r L^2}{8T} = \frac{2.2515 \times 275^2}{8 \times 3975}$$

$$= 5.3544 \text{ m.}$$

$$\text{Sag in vertical direction} = Sag \times \frac{w_c + w_i}{w_r}$$

$$= 5.3544 \times \frac{1.6391}{2.2515}$$

$$= 3.90 \text{ m}$$

Q.3. (a) In a six-unit suspensions insulator string, the capacitance between each link pin and earth is 0.1 of the self capacitance of each unit. Determine the voltage across the lowest unit as percentage of the total voltage and also the string efficiency. **(6.5)**

Ans.

$$\text{No of insulator units} = 6 = n$$

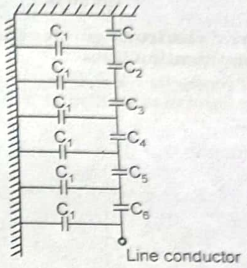
$$k = 0.1$$

$$C = 10\,C_1$$

$$C_2 = (1 + k)\,C = (1 + 0.1)\,C = 11\,C_1$$

$$C_3 = (1 + 3\,k)\,C = (1 + 0.1 \times 3)\,10\,C_1$$

$$= 13\,C_1$$



$$C_4 = (1 + 6k)\,C = (1 + 0.1 \times 6)\,10\,C_1$$

$$= 16\,C_1$$

$$C_5 = C_4 + 4\,kC = (1 + 6\,k)\,C + 4\,kC$$

$$= (1 + 10k)\,C = (1 + 10 \times 0.1)\,10\,C_1$$

$$= 20\,C_1$$

$$C_6 = (1 + 15\,k)\,C = (1 + 15 \times 0.5)\,10\,C_1$$

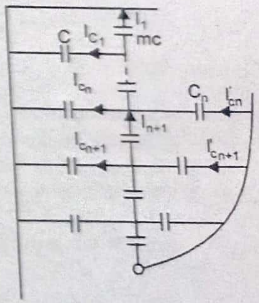
$$= 25\,C_1$$

Q.3. (b) Name methods of equalizing potential over string of insulator and explain method of static shielding. **(6)**

Ans. Method of equalizing potential are

(*i*) Selection of m (*ii*) Grading of units (*iii*) Static shielding.

Static Shielding: The idea is to cancel exactly the pin to tower charging currents so that the same current flows through the units of identical capacities to produce equal voltage drops across each unit. In this method a guard ring is connected round the power conductor such that it surrounds the bottom unit.



Since identical units are being used their mutual capacities are equal. Similarly the ground capacitance are equal. The design of the ring should be such that this gives rise to capacitances which cancel exactly the charging current in that particular section

$$I_{n+1} = I_n$$

and

$$I_{ca} = I'_{ca}$$

$$mV\omega C = (V - nv)\,\omega c_n$$

where

$$V = \text{operating voltage}$$

$$C_n = \text{Capacitance between guard ring and pin}$$

$$nV\omega C = (k - n)\,V\omega\,C_n$$

$$\Rightarrow \quad C_n = \frac{n}{k - n}\,C$$

Grading ring serves two purposes

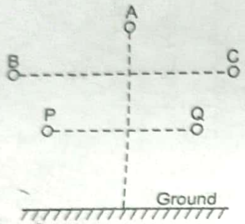
(*i*) equalisation of voltage drop across the string.

(*ii*) when used with arcing horn, it protects the insulator string from flashover.

UNIT-II

Q.4. (a) Explain in brief electromagnetic effect considering interference between power and communication lines. (6)

Ans. Considering power conductors A, B, C & two telephone conductors P and Q of a telephone line are running parallel to each other.



Let distances $d_{AP}, d_{AQ}, d_{BP}, d_{BQ}, d_{cp}, d_{cQ}$ respectively

$$L_{AP} = 0.2 \log_e \frac{d_{AP}}{r} \text{ mH/km}$$

$$L_{AQ} = 0.2 \log_e \frac{d_{AQ}}{r} \text{ mH/km}$$

$$M_A = L_{AQ} - L_{AP} = 0.2 \log_e \frac{d_{AP}}{r} - 0.2 \log_e \frac{d_{AQ}}{r}$$

$$= 0.2 \log_e \frac{d_{AQ}}{d_{AP}} \text{ mH/km}$$

similarly

$$M_B = 0.2 \log_e \frac{d_{BQ}}{d_{BP}} \text{ mH/km}$$

similarly

$$M_C = 0.2 \log_e \frac{d_{CQ}}{d_{CP}} \text{ mH/km}$$

$$M = M_A + M_B + M_C$$

$$= 0.2 \log_e \frac{d_{AQ}\, d_{BQ}\, d_{CQ}}{d_{AP}\, d_{BP}\, d_{CP}} \text{ mH/km}$$

$$V_m = 2\pi f MI \text{ volts/km.}$$

$$M = \text{mutual inductance.}$$

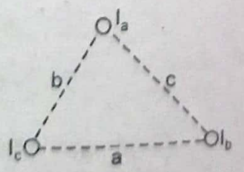
Q.4. (b) Derive a relation for average inductance per phase of unsymmetrical three phase line. (6.5)

Ans. Flux linkage of conductor a

$$\lambda_a = 2\times10^{-7}[I_a ln \frac{1}{R'} + I_b ln \frac{1}{C} + I_C ln \frac{1}{b}]$$

$$\lambda_b = 2\times10^{-7}\left[I_a ln \frac{1}{c} + I_b \frac{1}{R'} + I_c ln \frac{1}{a}\right]$$

$$\lambda_c = 2\times10^{-7}\left[I_a ln \frac{1}{b} + I_b ln \frac{1}{a} + I_c ln \frac{1}{R'}\right]$$



Now taking I_a as reference

$$I_b = k^2 I_a$$

$$I_c = k I_a$$

where $k = (-0.5 + j\,0.866)$

$$\lambda_a = 2\times10^{-7}\left[I_a ln \frac{1}{R'} + I_a(-0.5 - j0.866) ln \frac{1}{c} + I_a(-0.5 + j0.866) ln \frac{1}{b}\right]$$

$$= 2\times10^{-7} \times I_a\left[ln \frac{1}{R'} + (-0.5 - j0.866) ln \frac{1}{c} + (-0.5 + j0.866) ln \frac{1}{b}\right]$$

$$L_a = 2\times10^{-7}\left[ln \frac{1}{R'} - ln \frac{1}{\sqrt{bc}} - j\frac{\sqrt{3}}{2} ln \frac{b}{c}\right] = \frac{\lambda_a}{I_a}$$

Similarly

$$L_b = 2\times10^{-7}\left[ln \frac{1}{R'} - ln \frac{1}{\sqrt{ac}} - j\frac{\sqrt{3}}{2} ln \frac{c}{a}\right] = \frac{\lambda_b}{I_b}$$

$$L_c = 2\times10^{-7}\left[ln \frac{1}{R'} - ln \frac{1}{\sqrt{ab}} - j\frac{\sqrt{3}}{2} ln \frac{a}{b}\right] = \frac{\lambda_c}{I_c}$$

$$L = \frac{L_a + L_b + L_c}{3}$$

$$= \frac{1}{3}\left[2\times10^{-7}\left(3ln \frac{1}{R'} - ln \frac{1}{\sqrt{abc}} - j\frac{\sqrt{3}}{2} ln 1\right)\right]$$

$$= 2\times10^{-7} ln \frac{3\sqrt{abc}}{R'} \text{ Henry/meter}$$

For symmetrical spacing $a = b = c = d$

$$L = 2\times10^{-7} \ln \frac{d}{R'} \text{ Henry/meter.}$$

Q.5. (a) Explain analysis of Travelling wave with use of bewley diagram. (6.5)

Ans. This is a diagram which shows at a glance the position and direction of motion of every incident, reflected and transmitted wave on the system at every instant of time. Providing that the systems of lines is not too complex the difficulty of keeping track of the multiplicity of successive reflections is simplified.

Example let $R = 0.5\ \Omega/\text{km}, G = 10 \times 10^{-7}\ \text{S/km}, l = 400\ \text{km}$

Assumption $RC = GL$

A = Amplitude at any point of line

A_x = Amplitude at point distance x.

$$A_x = Ae^{-\alpha x}$$

For distortionless line $\alpha = \sqrt{RG}$

$$= \sqrt{0.5\times10\times10^{-7}} = 0.000707$$

when $x = l = 400\ \text{km},\ e^{-0.2828} = 0.7536$

{Source impedance = 0 — E, I incident waves with respect to the source}

$$\Gamma = \frac{0 - Z_{c1}}{Z_{c1} + 0} = -1$$

At receiving end line is open circuited and $\Gamma = 1$

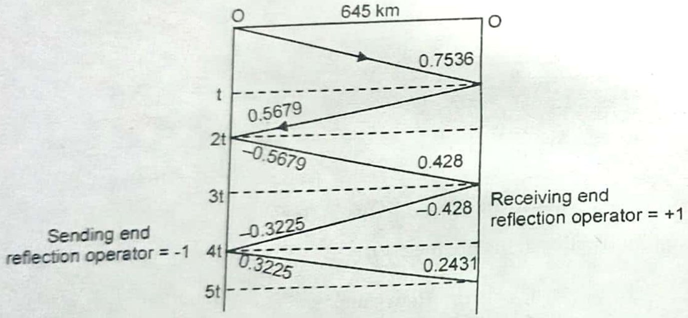
Let $V_s = 1\,Pu$, then we will have the following sequence of events as far as the reflected wave is concerned

$$t = 400/3\times10^5 = 0.00135$$

	Amplitude of wave	= 0.7536
At time 2t	Amplitude of wave	= 0.5679
	reflected wave	= − 0.5679
At time 3t	Amplitude of wave	= − 0.428
	reflected wave	= − 0.428
At time 4t	Amplitude of wave	= − 0.3225
	reflected wave	= + 0.3225 and so on.

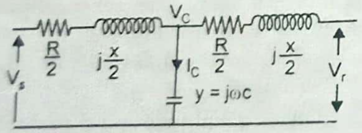
It is simpler to express in terms of α

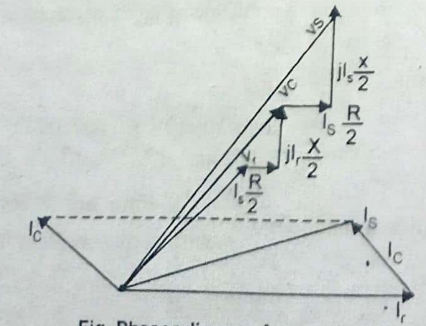
thus $$= 2(\alpha - \alpha^3 + \alpha^5 - \alpha^7 \ldots) = \frac{2\alpha}{1+\alpha^2}$$



Q.5. (b) For nomial T model of a medium transmission line deduce A, B, C, D parameters also explain its phasor diagram. (6)

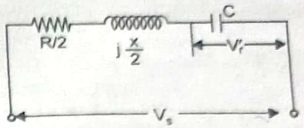
Ans.





Fig–Phasor diagram for nominal T.

Equivalent circuit under no load



$$V_c = (|V_r|\cos\phi_r + j|V_r|\sin\phi_r) + I_r\left(\frac{R}{2} + j\frac{X}{2}\right)$$

$$I_C = j\omega CV_c$$

$$I_s = I_c + I_r$$

$$= I_r + j\omega CV_c$$

$$V_s = V_c + I_s\left(\frac{R}{2} + j\frac{X}{2}\right)$$

$$= (|V_r|\cos\phi_r + j|V_r|\sin\phi_r) + I_r\left(\frac{R}{2} + j\frac{X}{2}\right) + I_s\left(\frac{R}{2} + j\frac{X}{2}\right)$$

$$V_r' = \frac{|V_s|(-j/\omega_c)}{\frac{R}{2} + j\frac{X}{2} - \frac{j}{\omega_c}}$$

$$\%\text{ regulation} = \frac{V_r' - V_r}{V_r}\times100$$

$$\%\eta = \frac{\text{Power delivered at receiving end}\times100}{\text{Power delivered at receiving end + loss.}}$$

$$V_c = V_r + I_r\frac{Z}{2}$$

$$I_c = V_c\,\alpha$$

$$I_s = I_r + I_c = I_r + V_c y = I_r + \left(V_r + I_r\frac{Z}{2}\right)y$$

$$V_s = V_c + I_s\frac{Z}{2} = V_r + I_r\frac{Z}{2} + \left\{I_r + \left(V_r + I_r\frac{Z}{2}\right)y\right\}\frac{Z}{2}$$

$$= V_r\left(1 + \frac{4Z}{2}\right) + I_r\left(Z + \frac{4Z^2}{4}\right)$$

$$I_s = I_r\left(1 + \frac{yZ}{2}\right) + V_r y = yV_r + \left(1 + \frac{yZ}{2}\right)I_r$$

$$\Rightarrow \quad A = 1 + \frac{yZ}{2}$$

$$B = Z\left(1 + \frac{yZ}{4}\right)$$

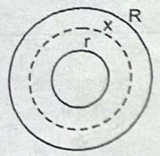
$$C = y$$

$$D = \left(1 + \frac{yZ}{2}\right)$$

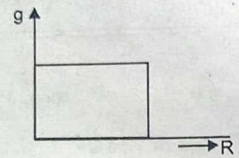
UNIT-III

Q.6. (a) Describe capacitance grading in cables to equalize the stress in the dielectric of cable. (6.5)

Ans. Capacitance Grading:



Capacitance grading



Grading with ∞ number of materials

Let λ = Charge/ length

If we have one single dielectric material, gradient will be

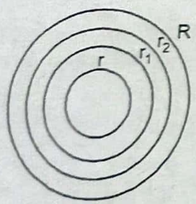
$$g = \frac{\lambda}{2\pi \in x}$$

where $\in$ = permittivity of material

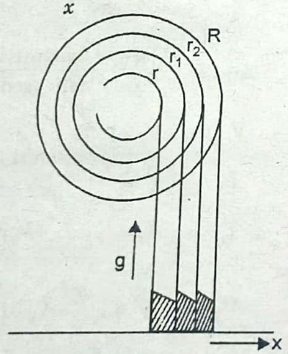
x = radius of material

$$\in = \frac{k}{x}$$

$$g = \frac{\lambda}{2\pi \frac{k}{x} . x} = \frac{\lambda}{2\pi k}$$



Single core cable with three materials



Capacitance grading voltage distribution

Let dielectric strength = G_1, G_2, G_3

Working stresses = g_1, g_2, g_3

Following possibilites are there

(i) Factor of safety for all the materials be same, thereby the working stress of various materials different.

$$\frac{\lambda}{2\pi \in_1 r} = \frac{G_1}{f}$$

where f = factor of safety.

The gradient at radius $r_1 = \dfrac{\lambda}{2\pi \in_2 r_1} = \dfrac{G_2}{f}$

The gradient at radius $r_2 = \dfrac{\lambda}{2\lambda \in_3 r_2} = \dfrac{G_3}{f}$

$$\Rightarrow \quad \lambda = 2\pi \in_1 r\frac{G_1}{f} = 2\pi \in_2 r_1 \frac{G_2}{f} = 2\pi \in_3 r_2 \frac{G_3}{f}$$

$$\in_1 r G_1 = \in_2 r_1 G_2 = \in_3 r_2 G_3$$

since $r < r_1 < r_2$

$$\in_1 G_1 > \in_2 G_2 > \in_3 G_3$$

(ii) The same working stress for different materialy

$$g_{max} = \frac{\lambda}{2\pi \in_1 r} = \frac{\lambda}{2\pi \in_2 r_1} = \frac{\lambda}{2\pi \in_3 r_2}$$

$$\Rightarrow \quad \in_1 r = \in_2 r_1 = \in_3 r_2$$

since $r < r_1 < r_2$

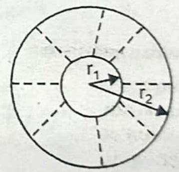
$$\in_1 > \in_2 > \in_3$$

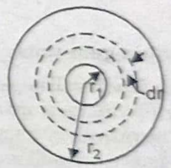
$$V = g_{max} r\, ln\frac{r_1}{r} + g_{max} r_1\, ln\frac{r_2}{r_1} + g_{max} r_2\, ln\frac{R}{r_2}$$

$$= g_{max}\left[r\, ln\frac{r_1}{r} + r_1\, ln\frac{r_2}{r_1} + r_2\, ln\frac{R}{r^2}\right]$$

Q.6. (b) What do you understand by insulation resistance? (6)

Ans. The cable conductor is provided with an insulation of suitable thickness so as to avoid leakage of current. The path for leakage current is radial through the insulating material. The opposition offered by the insulation to the leakage current is called the insulation resistance of the cable.





Insulation resistance offered to the leakage current by elementary cylinderical section of the insulation under consideration is $\dfrac{\rho dr}{2\pi r l}$

Insulation resistance of the cable

$$R_{INS} = \int_{r_1}^{r_2} \frac{\rho dr}{2\pi r l} = \frac{\rho}{2\pi l}[\log_e r]_{r_1}^{r_2}$$

$$= \frac{\rho}{2\pi l}\log_e \frac{r_2}{r_1}$$

Insulation resistance of the cable varies inversely as the length of the cable $\left(R_{INS} \alpha \frac{1}{l}\right)$

Q.7. (a) With a help of diagram and waveforms explain effect of sudden short circuit at the armature terminals of a three phase generator. (7.5)

Ans. Assumption

→ No dc off set in armature current

→ Magnitude of current decreases exponentially from a high initial value.

Instantaneous expression for fault current

$$I_f(t) = \sqrt{2}\, V_s \left[\left(\frac{1}{X_d''} - \frac{1}{X_d'} \right) e^{-t/T_d''} + \left(\frac{1}{X_d'} - \frac{1}{X_d} \right) e^{-t/T_d'} + \frac{1}{X_d} \right]$$

$$\sin\left(\omega t + \alpha - \frac{\pi}{2}\right)$$

where

V_s = Magnitude of terminal voltage

X_d'' = Direct axis subtransient reactance

X_d' = Direct axis transient reactance

X_d = Direct axis synchronous reactance

with $X_d'' < X_d' < X_d$. The time constants are

T_d'' = Direct axis substransient time constant

T_d' = Direct axis transient time constant.

Armature current of a synchronous generator as a short circuit occurs on its terminals.

$$I_f(0) = I_f'' = V_t / X_d''$$

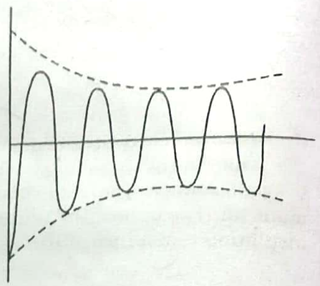
$$I_f' = V_t / X_d'$$

$$I_f = V_t / X_d$$

$$I_f = V_t / X_d$$

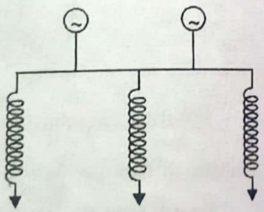
$$i_{dc}^{\max} = \sqrt{2}\, I_f'' e^{-t/T_A}$$

where T_A = armature time constant.

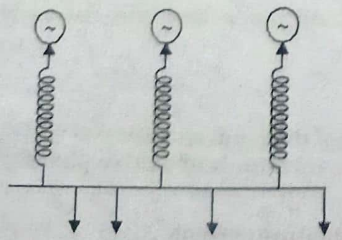


Q.7. (b) Explain generator reactors and feeder reactors. **(5)**

Ans. Feeder Ractor: The per unit value of reactance of a feeder based on its ratings may be small but when compared with the ratings of the whole system, its value is quite large and hence a small reactor will be effective in limiting the short circuit current should be a fault occur close to the generating station. In case this feeder reactor is not there, a fault in such a location would bring the bus bar voltage almost down to zero value and there is a possibility of various generators falling out of step.



Generator reactor: The reactance of modern alternator may be as high as 2.0 Pu which means even a dead short-circuit at the terminals of the alternator will result in a current less than full load current and therefore no external reactor is required for limiting the short circuit current of such a machine.



UNIT-IV

Q.8. (a) Describe Gauss-Sidedel method using Y-bus for power flow analysis. **(9.5)**

Ans. Gauss Siedel interative method

$$V_2^{k+1} = \frac{1}{V_{22}} \left[\frac{P_2 - jQ_2}{(V_2^k)^*} - (y_{21} Y_1 + Y_{23} V_3^k) \right]$$

in general

$$V_p^{k+1} = \frac{1}{Y_{pp}} \left[\frac{P_p - jQp}{(V_p^k)^*} - \sum_{\substack{q=1 \\ q \neq p}}^{n} Y_{pq} V_p^k \right]$$

where

$$B_{pq} = Y_{pq}$$

$$P = 1, 2, \ldots, n, P \neq S$$

The following points must be kept in mind

(*i*) Voltage and admittances are complex quantities.

(*ii*) Buses 2, 4 and 5 are load buses where P and Q are known quantities.

$$Q_p = \sum_{q=1}^{n} \{ f_p (e_q G_{pq} + f_q B_{pq}) - e_p (f_p G_{pq} - e_q B_{pq}) \}$$

where

$$y_{pq} = G_{pq} - j B_{pq}$$

$$V_p = e_p + j f_p$$

(*iii*) The number of non linear equation is $(n-1)$, where n is the number of buses in the system.

Procedure:

(*i*) Assume a flat voltage profile $1 + j_0$ for all nodal voltages except the slack bus.

(*a*) Set iteration Count $k = 0$

(*b*) Set bus count $p = 1$

(*c*) Check for slack bus. If it is slack bus, go to step 4 (a) otherwise go to next step.

(*ii*) Check which of the buses are voltage controlled and which are load buses. For voltage controlled, go to next statement otherwise go to step 4.

(*iii*) Replace the value of voltage magnitude of voltage controlled bus and keep same phase angle.

(*iv*) For bus p evaluate V_p^{k+1} and $\Delta V_p^k = V_p^{k+1} - V_p^k$

(*iv*) (*a*) Advance the bus count by 1 and check if all the buses have been taken into account. If yes go to next step otherwise go to 1(*c*)

(*v*) Find out the largest of absolute value of the change in voltage, if this is less than a prespecified tolerance, move on to the next step, otherwise go to step 2.

(*vi*) Calculate the injected powers and the line flows using the nodal voltages.

Q.8. (b) Derive static load flow equations (SLFE). **(3)**

Ans.

$$I_p = \sum_{q=1}^{n} Y_{pq} V_q \quad p = 1, 2, \ldots, n$$

$$I_p = Y_{pp} V_p + \sum_{\substack{q=1 \\ q \neq p}}^{n} Y_{pq} V_q$$

$$V_p = \frac{I_p}{Y_{pk}} - \frac{1}{Y_{pp}} \sum_{\substack{q=1 \\ q \neq p}}^{n} Y_{pq} V_q$$

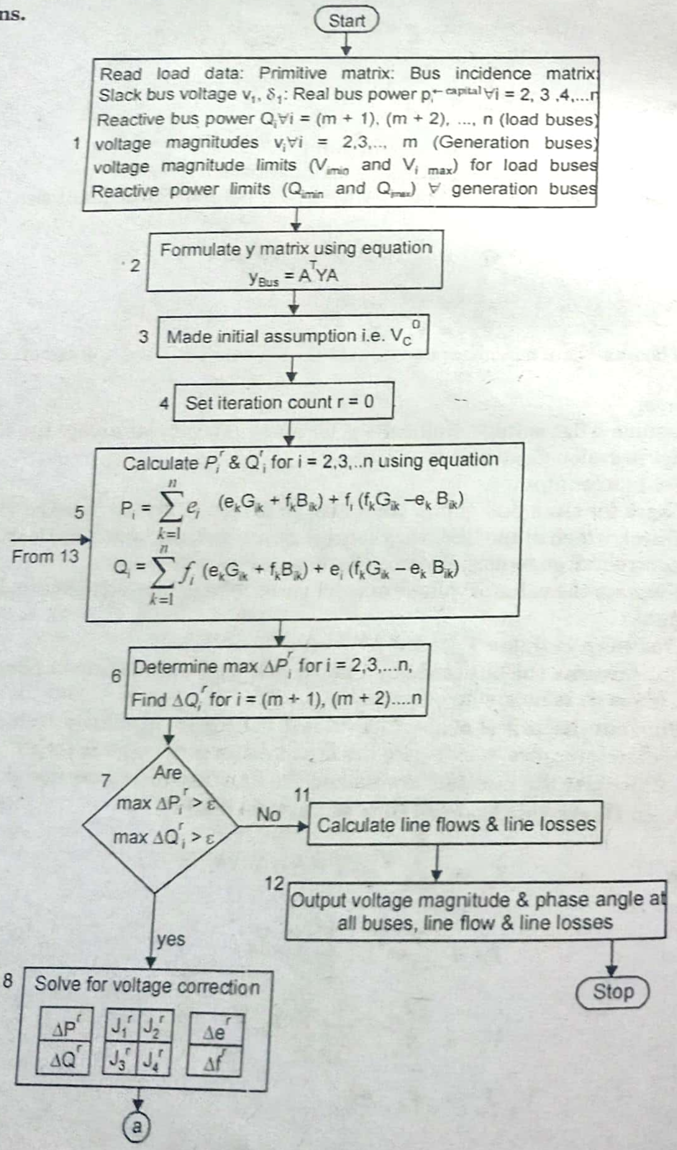
$$V_p^* I_p = P_p - jQ_p$$

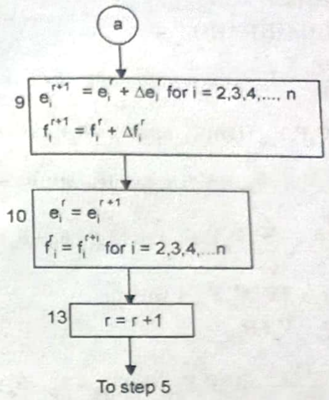
$$I_p = \frac{P_p - jQ_p}{V_p^*}$$

$$V_p = \frac{1}{Y_{pp}}\left[\frac{P_p - jQ_p}{V_p^*} - \sum_{\substack{q=1\\q\neq p}}^{n} Y_{pq}V_q\right] p = 1,2,...,n$$

Q.9. (a) Draw flow chart for load flow solution using Newton-Raphson method in polar coordinates. (8.5)

Ans.





Q.9. (b) Explain Fast Decoupled Load FLow (FDLF) method for analysis of load flow. (4)

Ans.

$$P_p - jQ_p = V_p^* I_p$$

$$I_p = \sum_{q=1}^{n} Y_{pq}V_q$$

$$\Rightarrow \quad P_p - jQ_p = V_p^* \sum_{q=1}^{n} Y_{pq}V_q$$

$$V_p = |V_p| e^{j\delta p}$$

$$Y_{pq} = |Y_{pq}| e^{-j\theta_{pq}}$$

$$P_p - jQ_p = |V_p| e^{-j\delta_p} \sum_{q=1}^{n} |Y_{pq}| e^{-j\theta_{pq}} |V_q| e^{j\delta_q}$$

$$= \sum_{q=1}^{n} |V_p||V_q||Y_{pq}| e^{j(-\delta_p + \delta_q - \theta_{pq})}$$

$$\Rightarrow \quad P_p = \sum_{q=1}^{n} |V_p||V_q||Y_{pq}| \cos(\theta_{pq} + \delta_p - \delta_q)$$

$$Q_p = \sum_{q=1}^{n} |V_p||V_q||Y_{pq}| \sin(\theta_{pq} + \delta_p - \delta_q)$$

$$P_p = \sum_{\substack{q=1\\q\neq p}}^{n} |V_pV_qY_{pq}| \cos(\theta_{pq} + \delta_p - \delta_q) + |V_p^2Y_{pp}| \cos\theta_{pp}$$

$$Q_p = \sum_{\substack{q=1\\q\neq p}}^{n} |V_pV_qY_{pq}| \sin(\theta_{pq} + \delta_p - \delta_q) + |V_p^2Y_{pp}| \sin\theta_{pp}$$

$$\begin{bmatrix}\Delta P\\ \Delta Q\end{bmatrix} = \begin{bmatrix}H & N\\ M & L\end{bmatrix}\begin{bmatrix}\Delta\delta\\ \Delta|E|/_{|E|}\end{bmatrix}$$

$$[\Delta P] = [H][\Delta\delta]$$

$$[\Delta Q] = [L][\Delta|E|/|E|]$$

$$H_{pq} = \frac{\partial P_p}{\partial \delta_q} = |V_p V_q Y_{pq}| \sin(\theta_{pq} + \delta_p - \delta_q)$$

$$= |V_p V_q Y_{pq}| \left(\sin\theta_{pq} \cos(\delta_p - \delta_q) + \sin(\delta_p - \delta_q) \cos\theta_{pq}\right)$$

$$= |V_p V_q| \left(-B_{pq} \cos(\delta_p - \delta_q) + G_{pq} \sin(\delta_p - \delta_q)\right)$$

$$H_{PP} = \frac{\partial P_p}{\partial \delta_p} = -\sum_{\substack{q=1 \\ q \neq p}}^{n} |V_p V_q Y_{pq}| \sin(\theta_{pq} + \delta_p - \delta_q)$$

$$= -Q_p + |V_p V_p Y_{pp}| \sin\theta_{pp}$$

$$= -Q_p - V_p^2 B_{pp}$$

$$L_{pq} = \frac{\partial Q_p |V_q|}{\partial V_q} = |V_p V_q Y_{pq}| \sin(\theta_{pq} + \delta_p - \delta_q)$$

$$= |V_p V_q| [-B_{pq} \cos(\delta_p - \delta_q) + G_{pq} \sin(\delta_p - \delta_q)]$$

$$L_{PP} = \frac{\partial Q_P |V_P|}{\partial V_p} = 2|V_p^2 Y_{pp}| \sin\theta_{pp} + \sum_{\substack{q=1 \\ q \neq p}}^{n} |V_p V_q Y_{pq}| \sin(\theta_{pq} + \delta_p - \delta_q)$$

$$= Q_p + |V_p^2 y_{pp}| \sin\theta_{pp}$$

$$= Q_p - V_p^2 B_{pp}$$

$$\cos(\delta_p - \delta q) \simeq 1$$

$$G_{pq} \sin(\delta_p - \delta_q) \leq B_{pq}$$

$$Q_p << B_{pp} V_p^2$$

$$L_{pq} = H_{pq} - |V_p V_q| B_{pq} \quad \forall\, q \neq P$$

$$L_{pp} = H_{pp} = -B_{pp} |V_p|^2$$

END TERM EXAMINATION [MAY-JUNE 2017]
FOURTH SEMESTER [B.TECH.]
POWER SYSTEM-I [ETEE-206]

Time : 3 hrs. **M.M. : 75**

Note: *Attempt any five questions including Q.no. 1 which is compulsory.*

Q.1. Attempt all of the following:

Q.1. (a) Prove that the per unit value of the 3 phase KVA on the 3 phase KVA base is identical to the per unit value of the KVA per phase on the KVA per phase base. **(5)**

Ans. In three phase system

Let KVA_B be the base value of KVA and kV_B be the base line to line voltage in KV

Assuming star connection

Base current $\quad I_B = \dfrac{kVA_B}{\sqrt{3}kV_B}$

Base impedance
$$Z_B = \frac{kV_B \times 1000}{\sqrt{3}I_B}$$
$$= \frac{\sqrt{3}kV_B}{kVA_B} \cdot \frac{kV_B}{\sqrt{3}}$$
$$= \frac{(kV_B)^2 \times 1000}{kVA_B}\Omega$$

Per unit impedance
$$Z_{pu} = \frac{\text{Actual impedance}}{\text{Base impedance}}$$
$$= \frac{\text{Actual impedance} \times kVA_B}{(kV_B)^2 \times 1000}$$

For a single phase circuit
$$S_B = V_B I_B$$
$$Z_B = V_B/I_B$$

Let kVA_B and kV_B are the base kVA and base kV.

Per unit $\quad kV = \dfrac{\text{Actual kV}}{\text{Base kV}} = \dfrac{\text{kV actual}}{kV_B}$

Base current $\quad I_B = \dfrac{\text{Base kVA}}{\text{Base kV}} = \dfrac{kVA_B}{kV_B}$

Per unit current $\quad I_{pu} = \dfrac{\text{Actual current}}{\text{Base current}} = \dfrac{\text{Actual current} \times kV_B}{kVA_B}$

Base impedance $\quad Z_B = \dfrac{\text{Base kV} \times 1000}{\text{Base current}} = kV_B \times \dfrac{kV_B}{kVA_B} \times 1000\Omega$

$$= \frac{(kV_B)^2 \times 1000}{kVA_B}$$

Same as in 3-ϕ system.

Q.1. (b) What are the requirements of the cables? **(3)**

Ans. An underground cable can be defined as the group of individually insulated one or more conductors which is put together and finally provided with number of layers of information to give support.

The basic necessary requirements of cables are

1. The size of conductor used must be such that is should carry the specified load without overheating and keeping the voltage drop well within the permissible limits.

2. At the voltage levels for which cables are designed, the insulation thickness must be proper so as to provide high degree of safety and the reliability.

3. The cables must be surrounded by number of layers of an additional insulation so as to give proper mechanical strength and protection. Thus the cables can with stand the rough use at the time of laying them.

4. The materials used in the manufacturing of cables must be such that there is complete chemical and physical stability throughout.

Q.1. (c) Why we neglect the load during the fault? Explain **(3)**

Ans. The fault current is the maximum current the transformer is capable of the only impedance considered for fault current is the transformer impedance.

Fault current is based on balted fault condition or Zero impedance.

Putting a load impedance in parallel with a zero impedance bolted fault still results in zero impedance under bolted fault conditions, the load voltage is zero and with zero voltage, the load current is zero.

Q.1. (d) What are the factors that minimize the corona effect? **(3)**

Ans. Corona effect can be reduced by following methods

(i) By increasing conductor size: By increasing conductor size, the voltage at which corona occurs is raised and hence corona effects are considerably reduced. This is one of the reasons that ACSR conductors which have a larger diameter are used in transmission lines.

(ii) By increasing conductor spacing: By increasing the spacing between conductors, the voltage at which corona occurs is raised and hence corona effect can be eliminated. However, spacing cannot be increased too much otherwise the cost of supporting structure may increase to a considerable extent.

Q.1. (e) How does the skin effect vary with conductor materials? **(3)**

Ans. In a good conductor, skin depth is inversely proportional to square root of resistivity. This means that better conductors have a reduced skin depth. The overall resistance of the better conductor remains lower even with reduced skin depth. However, the better conductor will show a higher ratio between its AC and DC resistance. Skin depth also varies as the inverse square root of permeability of the conductor.

The skin effect also reduces the effective thickness of laminations in power transformers, increasing their losses.

Q.1. (f) What are the merits and demerits of per-unit system? **(3)**

Ans. Advantages of pu system

(i) Calculations are simplified.

(ii) The characteristics of machines, when described in pu system are specified by almost the same number regardless of the ratings of the machines. Thus pu system provides a method of comparison.

(iii) This method is useful to eliminate ideal transformers as circuit components since the typical power system contains thousands of transformers, this is a non-trivial saving.

(iv) For circuits connected by transformer, pu system is particularly suitable.

Drawbacks:

(i) Some equations that hold in the unscaled cases are modified when scaled into pu. Factors such as $\sqrt{3}$ and 3 are removed or added by this method.

(ii) Equivalent circuits of the components are modified, making them somewhat more abstract, sometimes phase shifts that are clearly present in the unscaled circuit vanish in the pu system.

Q.1. (*g*) Discuss the various types of buses used in load flow studies? (5)

Ans. In a power system, each bus or node is associated with four quantities, real and reactive powers, $|V|, \delta$. In a load flow solution, two out of four quantities are specified and the remaining two are required to be obtained through solution of the equations. Buses are classified in following ways

(i) Load bus

(ii) Generator or voltage controlled bus

(iii) Slack, swing or reference bus.

Load bus	Quantities specified	Quantities to be obtained
Load bus	P,Q	$\|V\|, \delta$
Generator bus	P, $\|V\|$	Q,δ
Slack bus	$\|V\|, \delta$	P,Q

The phase angle of voltage at the slack bus is usually taken as reference. In the following analysis the real and reactive components of voltage at a bus are taken as independent variable for the load flow solution. i.e.

$$V_i \angle \delta_i = e_i + jf_i$$

where e_i and f_i are real and reactive component of voltage at ith bus.

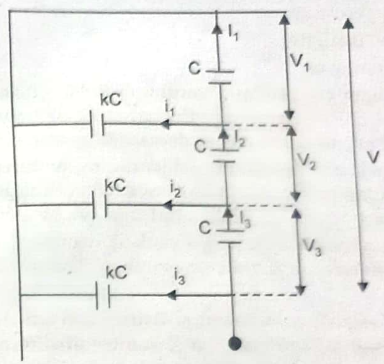
Q.2. (*a*) In a 33 kV overhead line, there are three units in the string of insulators. If the capacitance between each insulator pin and earth is 11% of self capacitance of each insulator, find (i) the distribution of voltages over 3 insulators and (ii) string efficiency (6.5)

Ans. $$K = \frac{\text{Shunt Capacitance}}{\text{Self Capacitance}} = 0.11$$

Voltage across string $$V = \frac{33}{\sqrt{3}} = 19.05 \text{ KV}$$

At junction A

$$\begin{aligned} I_2 &= I_1 + i_1 \\ V_2\omega c &= V_1\omega c + V_1 K\omega c \\ V_2 &= V_1(1+K) \\ &= 1.11\,V_1 \end{aligned}$$



At junction B

$$\begin{aligned} I_3 &= I_2 + i_2 \\ V_3\omega c &= V_2\omega c + (V_1+V_2)K\omega c \\ V_3 &= 1.11V_1 + (V_1 + 1.11\,V_1)0.11 \\ &= 1.342\,V_1 \end{aligned}$$

(*i*) Voltage across whole string is

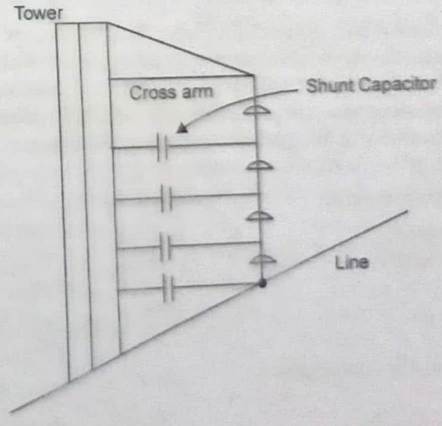
$$\begin{aligned} V &= V_1 + V_2 + V_3 \\ &= V_1 + 1.11V_1 + 1.342V_1 = 3.452\,V_1 \\ 19.05 &= 3.452\,V_1 \\ V_1 &= 5.52 \text{ KV} \\ V_2 &= 1.11V_1 = 6.13 \text{ KV} \\ V_3 &= 1.342\,V_1 = 7.5 \text{ KV} \end{aligned}$$

(*ii*) $$\text{String efficiency} = \frac{\text{Voltage across string}}{\text{No. of insulators} \times V_3} \times 100$$

$$= \frac{19.05}{3 \times 7.4} \times 100 = 85.8\%$$

Q.2. (*b*) What are methods of improving string efficiency? Explain (6)

Ans. Various methods of improving string efficiency are:



(i) By using longer cross arm

(ii) By grading the insulator

(iii) By using a guard ring

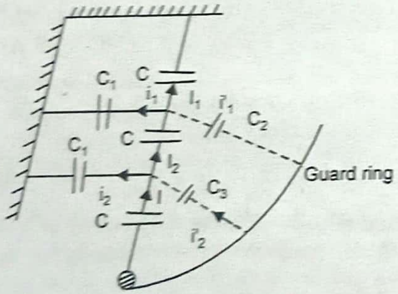
(i) By using a longer cross arm: The value of string efficiency depends on value of K. The lesser the value of K, the greater is the string efficiency and more uniform is the voltage distribution. The value of K can be decreased by reducing shunt capacitance.

In order to reduce shunt capacitance, distance of conductor from tower must be increased. However, limitations of cost and strength of tower do not allow the use of very long cross arms. In practice, K = 0.1 is the limit that can be achieved by this method.

(ii) By grading the insulators: In this method, insulators of different dimensions are so chosen that each has a different capacitance. The insulators are capacitance graded. Since $V \alpha 1/C$

This method tend to equalise the potential distribution across the units in the string. This method has the disadvantage that a large number of different sized insulators are required. However, good results can be obtained by using standard insulators for most of the string and larger units for that near to the line conductors.

(iii) By using a guard ring: The potential across each unit in a string can be equalised by using a guard ring which is a metal ring electrically connected to the conductor and surrounding the bottom insulator. The guard ring introduces capacitance between metal fittings and line conductor. The guard ring is contoured in such a way that i_1, i_2 etc are equal to metal fitting i'_1 and i'_2 etc. The result is that same charging current I flows through each unit of string. Consequently, there will be uniform potential distribution across the unit.



Q.3. Transmission line has a span of 275 m between level supports. The conductor has an effective diameter of 1.96 cm and weights 0.865 kg/m. Its ultimate strength is 8060 kg. If the conductor has ice coating of radial thickness 1.27 cm and is subjected to wind pressure of 3.9 gm/cm² projected area, calculate sag for a safety factor of 2. Weight of 1 c.c. ice is 0.91 gm. (12.5)

Ans. Span length l = 275 m

Wt. of conductor/m length, w = 0.865 kg

Conductor diameter d = 1.96 cm

Ice coating thickness, t = 1.27 cm

Working tension $T = \dfrac{8060}{2} = 4030$ kg

Volume of ice/m length of conductor

$$= \pi t(d+t) \times 100 \text{ cm}^3$$
$$= \pi \times 1.27\,(1.96 + 1.27) \times 100$$
$$= 1288 \text{ cm}^3$$

Weight of ice/m length of conductor = 0.91 × 1288

$$= 1172 \text{ gm} = 1.172 \text{ kg}$$

Wind force/m length of conductor is

$$w_w = \text{Pressure} \times (d + 2t) \times 100$$
$$= 3.9 \times (1.96 + 2 \times 1.27) \times 100$$
$$= 1755 \text{ gm} = 1.755 \text{ kg}$$

Total weight of conductor/m is

$$w_t = \sqrt{(w + w_i)^2 + w_w^2}$$
$$= \sqrt{(0.865 + 1.172)^2 + 1.755^2}$$
$$= 2.688 \text{ kg.}$$

$$\text{Sag} = \frac{w_t l^2}{8T} = \frac{2.688 \times 275^2}{8 \times 4030} = 6.3 \text{ m.}$$

Q.4. (a) What is CORONA? How it affects transmission lines? What are the factors affecting CORONA? What are the practical method to reduce CORONA? (6.5)

Ans. When an alternating potential difference is applied across two conductors whose spacing is large as compared to their diameters, there is no apparent change in the condition of atmospheric air surrounding the wires if the applied voltage is low. However, when the applied voltage exceeds a certain value called critical disruptive voltage, the conductors are surrounded by a faint violet glow called Corona.

The phenonmenon of corona is accompanied by a hissing sound, production of ozone, power loss and radio interference. The higher the voltage is raised, the larger and higher the luminous envelop becomes and greater are the sound, the power loss and radio noise. If the applied voltage is increased to breakdown value, a flash over will occur between the conductors due to breakdown of air insulation.

Factors affecting corona are:

(i) Atmosphere: As corona is formed due to ionisation of air surrounding the conductors, therefore it is affected by the physical state of atmosphere.

(ii) Conductor size: Corona effect depends upon the shape and conditions of conductors. Rough and irregular surface will give rise to more corona because uneveness of surface decreases value of breakdown voltage.

(iii) Spacing between conductors: If spacing between the conductors is made very large as compared to their diameters, there may not be any corona effect. It is because larger distances between conductors reduces electro-static stresses at the conductor surface, thus avoiding corona formation.

(iv) Line voltage: If line voltage is low, there is no change in the condition of surrounding air and hence no corona is formed.

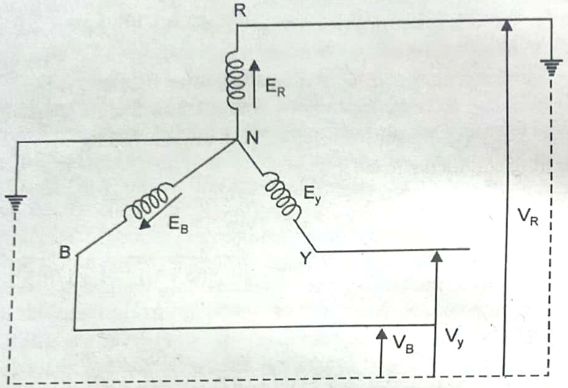
Methods of reducing corona losses are:

(i) By increasing conductor size

(ii) By increasing conductor spacing

Q.4. (b) Derive an expression for the fault current for a single line to ground fault. (6)

Ans.



$$\bar{V}_R = 0 \quad \bar{I}_B = \bar{I}_y = 0$$

Sequence currents in R phase in terms of line currents

$$\bar{I}_0 = \frac{1}{3}(\bar{I}_R + \bar{I}_y + \bar{I}_B) = \frac{1}{3}\bar{I}_R$$

$$\bar{I}_1 = \frac{1}{3}(\bar{I}_R + \alpha\bar{I}_y + \alpha^2\bar{I}_B) = \frac{1}{3}\bar{I}_R$$

$$\bar{I}_2 = \frac{1}{3}(\bar{I}_R + \alpha^2\bar{I}_y + \alpha\bar{I}_B) = \frac{1}{3}\bar{I}_R$$

$$\bar{E}_R = \bar{I}_1\bar{Z}_1 + \bar{I}_2\bar{Z}_2 + \bar{I}_0\bar{Z}_0 + \bar{V}_R$$

As $\quad \bar{V}_R = 0$ and $\bar{I}_1 = \bar{I}_2 = \bar{I}_0$

$$\bar{E}_R = \bar{I}_0(\bar{Z}_1 + \bar{Z}_2 + \bar{Z}_0)$$

$$\bar{I}_0 = \frac{\bar{E}_R}{\bar{Z}_1 + \bar{Z}_2 + \bar{Z}_0}$$

Fault current

$$\bar{I}_R = 3\bar{I}_0 = \frac{3\bar{E}_R}{\bar{Z}_1 + \bar{Z}_2 + \bar{Z}_0}$$

Q.5. (a) A 3 phase line has conductors 2 cm in diameter spaced equilaterally 1 m apart. If the dielectric strength of air is 30 KV (max)/cm, find the disruptive critical voltage for the line. Take the air density factor $\delta = 0.952$ and irregularity factor $m_0 = 0.9$. (6.5)

Ans. Conductor radius $\quad r = 2/2 = 1$ cm

Conductor spacing $\quad d = 1\text{m} = 100$ cm

Dielectric strength of air $\quad g_0 = 30$ kV/cm (max) $= 21.2$ kV/cm (rms)

Disruptive critical voltage

$$V_C = m_0 g_0 \delta r \log_e (d/r)$$

$$= 0.9 \times 21.2 \times 0.952 \times 1 \times \log_e\left(\frac{100}{1}\right)$$

$$= 83.64 \text{ kV/phase}$$

Line voltage (rms)

$$= \sqrt{3} \times 83.64 = 144.8 \text{ kV.}$$

Q.5. (b) What is the effect of load power factor on regulation and efficiency in a transmission line? (6)

Ans. Effect of load power factor on regulation: Expression for voltage regulation of a short transmission line is given by

$$\%\text{ voltage regulation (lagging pf)} = \frac{IR\cos\phi_R + IX_L\sin\phi_R}{V_R} \times 100$$

$$\%\text{ voltage regulation (leading pf)} = \frac{IR\cos\phi_R - IX_L\sin\phi_R}{V_R} \times 100$$

Following conclusion can be drawn

(i) When load pf is lagging or unity or such leading that IR $\cos\phi_R >$ IX_L $\sin\phi_R$, then V.R. is positive.

(ii) For a given V_R and I, V.R. of the line increases with the decrease in pf for lagging loads.

(iii) When load pF is leading to this extent that $I_0X_L \sin\phi_R >$ IR $\sin\phi_R$ the V.R. is negative

(iv) For a given V_R and I, V.R. of line decreases with decrease in pF for leading loads.

Effect of transmission line:

$$P_{1-\phi} = V_R{*}I\cos\phi_R \quad \text{(For 1–}\phi\text{ line)}$$

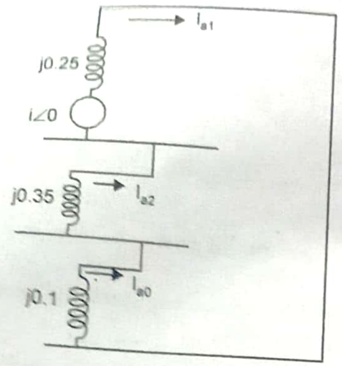
$$P_{3-\phi} = 3V_R I\cos\phi_R \quad \text{(For 3–}\phi\text{ line)}$$

$$I = \frac{P}{V_R\cos\phi_R} \quad \text{(For 1–}\phi\text{ line)}$$

$$I = \frac{P}{3V_R\cos\phi_R} \quad \text{(For 3–}\phi\text{ line)}$$

$$= I \propto \frac{1}{PF}$$

Q.6. A salient pole generator without damper is rated at 20 MVA, 13.8 KV and has a direct axis sub transient reactance of 0.25 pµ. The negative and zero sequence reactance are respectively 0.35 pm and 0.10 pµ. The neutral of the generator is solidly grounded. Determine the sub transient current in the generator and line-to-line voltages for sub transient conditions when single line to ground fault occur at the generator terminals with generator operating unloaded at rated voltage. Neglect resistance. (12.5)



$$S_b = 20\text{ MVA}$$
$$V_{base} = 13.8\text{ kV}$$
$$V_{base}ph = \frac{13.8}{\sqrt{3}}\text{kV}$$

Since the generator is unloaded

$$E_a = E_g = V_i = 1\angle 0$$

For an LG fault

$$I_{a1} = I_{a2} = I_{ao}$$
$$= \frac{E_a}{Z_1 + Z_2 + Z_0} = \frac{1\angle 0}{j0.1 + j0.35 + j0.25}$$
$$= -j1.428\text{ pu}$$
$$I_f = I_a = 3I_{a1} = -j4.29\text{ pu}$$
$$I_b = I_c = 0$$
$$I_{base} = \frac{20}{\sqrt{3}\times 13.8} = 0.837\text{ kA}$$
$$I_f = -j4.29 \times 0.837 = -j3.59\text{ kA}$$

The symmetrical component of voltage from point a to ground are

$$V_{a1} = Ea - I_{a1}Z_1 = 1-(-j1.428)\times j0.25$$
$$= 0.6425\text{ pu}$$
$$V_{a2} = -I_{a2}Z_2 = -(-j1.428)\times j0.35 = -0.5\text{ pu}$$
$$V_{a0} = -I_{a0}Z_o = -(-j1.428)\times j0.1 = -0.1425\text{ pu}$$

L – G voltages are

$$V_a = 0$$
$$V_b = V_{a0} + a^2V_{a1} + aV_{a2}$$
$$= -0.1425 + 0.6425\angle 240 - 0.5\angle 120$$
$$= -0.2143 - j0.9894\text{ pu}$$

$$V_C = V_{a0} + aV_{a1} + a^2V_{a2}$$
$$= -0.1425 + 0.6425\angle 120 - 0.5\angle 240°$$
$$= -0.2143 + j0.9894\text{ pu}$$

Line to line voltages

$$V_{ab} = V_a - V_b = 0.2143 + j0.9894 = 1.012\angle 77.780$$
$$V_{bc} = V_b - V_c = j1.9788 = 1.9788\angle -90°$$
$$V_{ca} = V_c - V_a = -0.2143 + j0.9894$$
$$= 1.012\angle 102.22°$$
$$V_{ab} = 1.012\angle 77.78 \times \frac{13.8}{\sqrt{3}} = 8.06\angle 77.78°\text{ kV}$$
$$V_{bc} = 1.9788\angle -90° \times \frac{13.8}{\sqrt{3}} = 15.764\angle -90°\text{ kV}$$
$$V_{ca} = 01.012\angle 102.22 = 8.06\angle 102.22°\text{ kV}$$

Q.7. One line diagram of a simple 4-bas system is given in fig. The relevant per unit line impedances are indicated on the diagram. The shunt admittanes at the buses may be neglected.

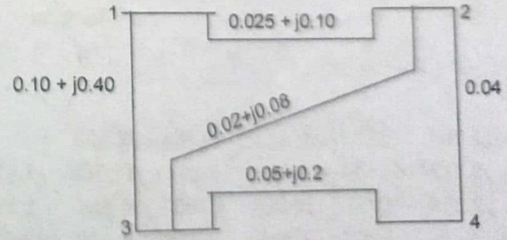
(a) Determine Y_{bus} if buses 1 and 2 are not connected, as indicated by dotted line in fig.

(b) What modifications will have to be carried out in Y_{bus} if the fuses 1 and 2 are connected as indicated by dotted lines in fig.

Ans. The line admittances are worked out as below:

$$y_{12} = \frac{1}{0.025 + j0.1} = \frac{0.025 - j0.1}{0.010625} = 2.353 - j9.412$$

$$y_{23} = \frac{1}{0.02 + j0.08} = \frac{0.02 - j0.08}{0.0068} = 2.941 - j11.765$$



$$y_{13} = \frac{1}{0.10 + j0.40} = \frac{0.10 - j0.40}{0.17} = 0.588 - j2.353$$

$$y_{34} = \frac{1}{0.05 + j0.2} = \frac{0.05 - j0.2}{0.0425} = 1.176 - j4.706$$

$$y_{24} = \frac{1}{0.04 + j0.16} = \frac{0.04 - j0.16}{0.0272} = 1.471 - j5.882$$

$y_{10} = y_{20} = y_{30} = y_{40} = 0$ because shunt adminttanus at buses are negligible.

When bases 1 an 2 are not connected, as shown by dotted lines, i.e. $y_{12} = 0$

$$Y_{bus} = \begin{bmatrix} y_{10}+y_{12}+y_{13} & -y_{12} & -y_{13} & 0 \\ 0 & y_{20}+y_{12}+y_{23}+y_{24} & -y_{23} & -y_{24} \\ -y_{13} & -y_{23} & y_{30}+y_{13}+y_{23}+y_{34} & -y_{34} \\ 0 & -y_{24} & -y_{34} & y_{40}+y_{24}+y_{34} \end{bmatrix}$$

$$= \begin{bmatrix} y_{13} & 0 & -y_{13} & 0 \\ 0 & y_{23}+y_{24} & -y_{23} & -y_{24} \\ -y_{13} & -y_{23} & y_{13}+y_{23}+y_{34} & -y_{34} \\ 0 & -y_{24} & -y_{34} & y_{24}+y_{34} \end{bmatrix}$$

$$Y_{bus} = \begin{bmatrix} 0.588 - j2.353 & 0 & -0.588 + j2.353 & 0 \\ 0 & 4.412 - j17.647 & -2.941 + j11.765 & -1.471 + j5.882 \\ -0.588 + j2.353 & -2.941 + j11.765 & 4.705 - j18.824 & -1.176 + j4.706 \\ 0 & -1.471 + j5.882 & -1.176 + j4.706 & 2.647 - j10.588 \end{bmatrix}$$

When buses 1 and 2 are connected, as shown by dotted line, i.e. when

$$y_{12} = 2.353 - j9.412$$

$$y_{bus} = \begin{bmatrix} y_{12}+y_{13} & -y_{12} & -y_{13} & 0 \\ -y_{12} & y_{12}+y_{23}+y_{24} & -y_{23} & -y_{24} \\ -y_{13} & -y_{23} & y_{13}+y_{23}+y_{34} & -y_{34} \\ 0 & -y_{24} & -y_{34} & y_{24}+y_{34} \end{bmatrix}$$

$$\text{or } Y_{bus} = \begin{bmatrix} 2.941 - j11.765 & -2.353 + j9.412 & -0.588 + j2.353 & 0 \\ -2.353 + j9.412 & 6.765 + j27.059 & -2.941 + j11.765 & -1.471 + j5.882 \\ -0.588 + j2.353 & -2.941 + j11.765 & 4.705 - j18.824 & -1.176 + j4.706 \\ 0 & -1.471 + j5.882 & -1.176 + j4.206 & 2.647 + j10.533 \end{bmatrix}$$

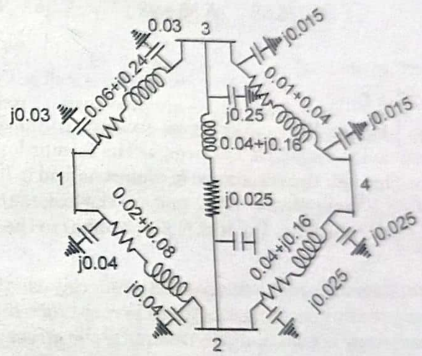
Q.7. (b) A sample power syster has the following line data, from bus admittance matrix for this system.

Bus code	Series Impedance	PU line charging Admittance y/2
1–2	0.02 + j0.08	0.0 + j0.04
1–3	0.06 + j0.24	0.0 + j0.03
2–3	0.04 + j0.16	0.0 + j0.025
2–4	0.04 + j0.016	0.0 + j0.025
3–4	0.01 + j0.04	0.0 + j0.015

Ans. The network is show in fig.

The line series admittance are worked-out as below:

$$y_{12} = \frac{1}{0.02 + j0.08} = \frac{0.02 - j0.08}{0.0068} = (2.94 - j11.76)\text{PU}$$



$$y_{13} = \frac{1}{0.06 + j0.24}$$

$$= \frac{0.06 - j0.24}{0.0612} = (0.98 - j3.92)\text{ pu}$$

$$y_{23} = \frac{1}{0.04 + j0.16}$$

$$= \frac{0.04 - j0.16}{0.0272} = (147 - j5.88)\text{ pu}$$

$$y_{24} = y_{23} = (1.47 - j5.88)\text{ pu}$$

$$y_{34} = \frac{1}{0.01 + j0.04} = \frac{0.01 - j0.04}{0.0017} = (5.88 - j23.53)\text{ pu}$$

The shunt admittances at different buses are worked out as below:

$$y_{10} = j0.03 + j0.04 = j0.07\text{ pu}$$

$$y_{20} = j0.04 + j0.025 + j0.025 = j0.09 \text{ pu}$$

$$y_{30} = j0.03 + j0.025 + j0.025 = j0.07 \text{ pu}$$

$$y_{40} = j0.025 + j0.015 = j0.04 \text{ pu}$$

$$Y_{bus} = \begin{bmatrix} y_{12}+y_{13}+y_{10} & -y_{12} & -y_{13} & 0 \\ -y_{12} & y_{12}+y_{13}+y_{24}+y_{20} & -y_{23} & -y_{24} \\ -y_{13} & -y_{23} & y_{13}+y_{23}+y_{34}+y_{30} & -y_{34} \\ 0 & -y_{24} & y_{34} & y_{24}+y_{34}+y_{40} \end{bmatrix}$$

$$= \begin{bmatrix} 3.92-j15.61 & -2.94+j11.76 & -0.98+j3.92 & 0 \\ -2.94+j11.76 & 5.88-j23.43 & -1.47+j5.88 & -1.47+j5.88 \\ -0.98+j3.92 & -1.47+j5.88 & 8.33-j33.26 & -5.88+j23.53 \\ 0 & -1.47+j5.88 & -5.88+j23.53 & 7.35-j29.37 \end{bmatrix}$$

Q.8. Write short notes on:

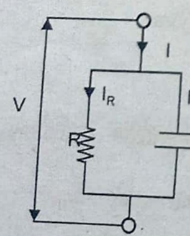
Q.8. (a) Dielectric Loss **(6)**

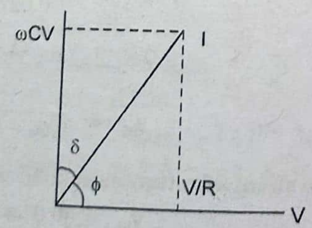
Ans. Dielectric Losses: The energy losses occuring in the dielectric of cables are due to leakage and so called dielectric hysteresis. The former loss is due to passing of current by conduction through the resistance of dielectric and is independent of supply. Frequency and therefore, it occurs both with dc and ac. The leakage current is proportional to the applied voltage and therefore, the loss is propotional to the square of the applied voltage.

With alternating voltages there is a further loss of energy, usually known as hystersis loss and this loss under usually operating conditions is very much higher than the leakage loss. Apparently some energy is consumed in reversing the stresses in solid dielectrics, and this appears as heat, causing increase in temperature of the dielectric. At normal operating stresses the dielectric hysteresis loss is proportional to the square of the electrostatic stresses, that is to the square of the applied voltage.

In order to take into account the above losses the charging current of a cable I is assumed to have two components:

One being true capacitance current which is equal to ωCV and leads the applied voltage by 90° and the other being the energy component which is in phase with the applied voltage and represents the dielectric loss component of current. The equivalent circuit for this system is represented by a parallel combination of leakage resistance R representing dielectric power loss and a capacitance C. The phasor diagram is shown





If V is the applied voltage C is the capacitance of cables, ϕ is the phase angle between voltage and current called the power factor angle of the cable and δ is the loss angle of the dielectric

Charging current $$I_C = \frac{V}{X_C} = \frac{V}{1/\omega C} = \omega CV$$

Dielectric loss per phase $$= VI\cos\phi = VI\sin\delta$$

$$= V\frac{I_c}{\cos\delta}\sin\delta$$

$$= V\omega CV\tan\delta$$

$$= \omega CV^2\tan\delta$$

Since δ is normally very small, so tan δ can be take8n to equal to δ and dielectric loss per phase is given by the expression

$$P = V^2\omega C\delta \text{ watts}$$

Q.8. (b) Ferranti effect **(6.5)**

Ans. Ferranti Effect: When a long line is operating under no load or light load condition the receiving end voltage is greater than the sending end voltage. This is known as ferranti effect. This phenomenon can be explained with the following reasonings:

(i) Assume no load condition.

$$V = \frac{V_r + I_r Z_C}{2} e^{\alpha x} e^{j\beta x} + \frac{V_r + I_r Z_C}{2} e^{-\alpha x} e^{-j\beta x}$$

reduces to

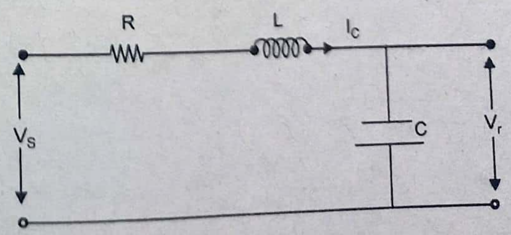
$$V_S = \frac{V_r}{2} e^{\alpha l} e^{j\beta l} + \frac{V_r}{2} e^{-\alpha l} e^{-j\beta l}$$

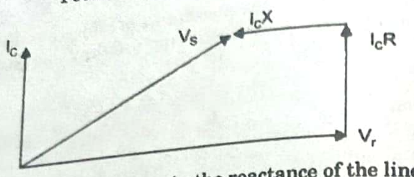
When $x = l$ and $I_r = 0$

At $l = 0$ $$V_r = \frac{V_r}{2} + \frac{V_r}{2}$$

As l increases, the incident component of sending end voltage increases exponentially and turns the vector anti clockwise through an angle βl whereas the reflected part of sending end voltage decreases by the same amount and is rotated clockwise through the same angle βl. The sum of these two component of sending end voltage gives a voltage which is smaller than V_r.

(*ii*) A simple explanation of Ferranti effect can be given by approximating the distributed parameters of the line by lumped impedance as shown in fig.





The charging current produces drop in the reactance of the line which is in phase opposition to the receiving end voltage and hence the sending end voltage becomes smaller than the reciving end voltage.

END TERM EXAMINATION [MAY-JUNE 2018]
FOURTH SEMESTER [B.TECH]
POWER SYSTEM-I [ETEE-206]

Time : 3 hrs. **M.M. : 75**

Note: *Attempt any five questions including Q No.1 which is compulsory. Assume missing data, if any.*

Q.1. (*a*) Write advantages of bundle conductors. **(5)**

Ans. Bundled conductors are primarily employed to reduce the corona loss and radio interference. However they have several advantages:

- Bundled conductors per phase reduces the voltage gradient in the vicinity of the line. Thus reduces the possibility of the corona discharge. (Corona effect will be observed when the air medium present between the phases charged up and start to ionize and acts as a conducting medium. This is avoided by employing bundled conductors)
- Improvement in the transmission efficiency as loss due to corona effect is countered.
- Bundled conductor lines will have higher capacitance to neutral in comparison with single lines. Thus they will have higher charging currents which helps in improving the power factor.
- Bundled conductor lines will have higher capacitance and lower inductance than ordinary lines they will have higher Surge Impedance Loading (Z=(L/C)1/2). Higher Surge Impedance Loading (SIL) will have higher maximum power transfer ability.
- With increase in self GMD or GMR inductance per phase will be reduced compared to single conductor line. This results in lesser reactance per phase compared to ordinary single line. Hence lesser loss due to reactance drop.

Q.1. (*b*) What is visual critical voltage in corona loss? **(5)**

Ans. Visual critical voltage

It is the minimum phase-neutral voltage at which **corona glow** appears all along the line conductors.

It has been seen that in case of parallel conductors, the corona glow does not begin at the disruptive voltage V_c but at a higher voltage V_v, called **visual critical voltage.**

The phase-neutral effective value of visual critical voltage is given by the following empirical formula :

$$V_V = m_v g_0 \delta_r \left(1 + \frac{0.3}{\sqrt{\delta_r}}\right) \log_e \frac{d}{r} \text{kV / phase}$$

where m_v is another irregularity factor having a value of 1·0 for polished conductors and 0·72 to 0·82 for rough conductors.

Q.1. (c) Explain effect of short circuit at the armature terminals of a 3-φ generator. **(5)**

Ans. In abnormal operation, the alternator may be subjected to transient conditions which cannot be explained from steady-state theory. These transients may occur from-(i) Switching (ii) Sudden changes of load, (iii) Sudden short circuit (either line-to-line or line-to-neutral or from a symmetrical short circuit).

Effects of short circuit currents can be conveniently studied by using the concept of constant flux linkage, which may be stated as below:

It is impossible to change the flux linkages in a closed electrical circuit that contains no resistance or capacitance (i.e. in a purely inductive circuit).

In case of a synchronous machine, the armature and field windings can be assumed almost purely inductive because they do not contain any capacitance and their resistances are almost negligible in comparison to their inductive reactances.

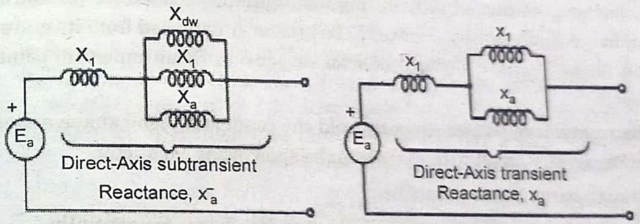
So the flux linkages in the armature circuit and the field circuit cannot be changed suddenly by the application of short circuit to the armature winding. For maintaining these flux linkages constant, large changes of current may take place in both of the windings when the short circuit occurs in order to keep their respective flux linkages constant.

Reactance of Synchronous Machines:

The current flowing in the armature of a synchronous generator when its terminals are short circuited is similar to that flowing when a sinusoidal voltage is suddenly applied to an R-L series circuit. However, there is one important difference, that is, in case of an R-L series circuit, reactance X (ωL) is a constant quantity where as in case of the synchronous generator the reactance is not a constant one but is a function of time.

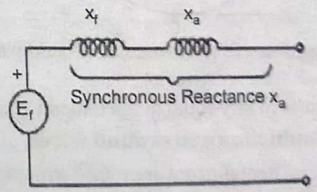
In practice three discrete values are assigned and thus we have three reactances-direct-axis subtransient reactance symbolised as X'_d, direct-axis transient reactance symbolised as X'_d and direct-axis synchronous reactance symbolized as X_d.

In short circuit studies only the above direct-axis reactances are involved. This can be justified as follows:



(a) Approximate Circuit. Model During Subtransient Period of Short Circuit

(b) Approximate Circuit Model During Transient Period of Short Circuit



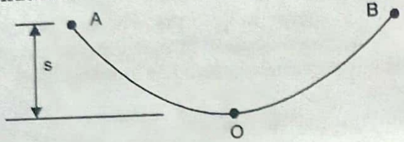
(c) Steady-State Short-Circuit Model of a Synchronous Machine

The reactance represented by the machine in the initial period of the short circuit i.e. $[X_l + \{1/(1/X_a) + (1/X_f) + (1/X_{dw})\}]$ is called the subtransient reactance (X''_d) of machine; while the reactance effective after the damper winding currents have died out, i.e., $[X_l + \{1/(1/X_a) + (1/X_f)\}]$ is called the transient reactance X'_d of the machine.

Of course, the reactance under steady-state short circuit conditions is the synchronous reactance of the machine. Obviously $X''_d < X'_d < X_d$. The machine thus offers a time-varying reactance which changes from X''_d to X'_d and finally to X_d.

Q.1. (d) Describe Deflected sag and factor and safety in over head lines. (5)

Ans. Sag is defined as the different in level between points of supports and the lowest point on the conductor.



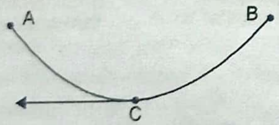
Here AOB is the transmission line conductor. Two supports are at point A and at point B. AB is the horizontal line and from this horizontal line to point O, S is the sag when measured vertically.

Why Sag Provision is Mandatory in Transmission Line Conductors? Sag is mandatory in transmission line conductor suspension. The conductors are attached between two supports with perfect value of sag. It is because of providing safety of the conductor from not to be subjected to excessive tension. In order to permit safe tension in the conductor, conductors are not fully stretched; rather they are allowed to have sag. If the conductor is stretched fully during installation, wind exerts pressure on the conductor, hence conductor gets chance to be broken or detached from its end support. Thus sag is allowed to have during conductor suspension. Some important points are to be mentioned:

1. When same leveled two supports hold the conductor, bend shape arises in the conductor. Sag is very small with respect to the span of the conductor.

2. Sag span curve is like parabolic.

3. The tension in each point of the conductor acts always tangentially.



4. Again the horizontal component of the tension of conductor is constant throughout conductor length.

5. The tension at supports is nearly equal to the tension at any point of the conductor.

Factors of safety of conductors and ground wires

The factor of safety (f.o.s) of a conductor (or ground wire) is the ratio of the ultimate strength of the conductor (or ground wire) to the load imposed under assumed loading condition. Rule 76 (1)(c) of the Indian Electricity Rules,

The minimum factor of safety for conductors shall be two, based on their ultimate tensile strength. In addition, the conductor tension at 32°C without.

Q.1. (e) Elaborate Bewley's Lattice diagram for analysis of travelling wave. (5)

Ans. Bewley Lattice Diagram: This is a convenient diagram devised by Bewley, which shows at a glance the position and direction of motion of every incident, reflected, and transmitted wave on the system at every instant of time. The diagram overcomes the difficulty of otherwise keeping track of the multiplicity of successive reflections at the various junctions.

Consider a transmission line having a resistance **r**, an inductance **l**, a conductance **g** and a capacitance **c**, all per unit length.

If γ is the propagation constant of the transmission line, and **E** is the magnitude of the voltage surge at the sending end, then the magnitude and phase of the wave as it reaches any section distance **x** from the sending end is **Ex** given by.

$$E = E.e = E.e = Eee - x - (+j)\, x - x - j\, xx\ \gamma\ \alpha\beta\ \alpha\ \beta$$

where

e - represents the attenuation in the length of line x

e^{-j} - represents the phase angle change in the length of line xTherefore, attenuation constant of the line in neper/km phase angle constant of the line in rad/km.

It is also common for an attenuation factor k to be defined corresponding to the length of a particular line. i.e.

k = e for a line of length l.

In the Bewley lattice diagram, the following properties exist.

- All waves travel downhill, because time always increases.
- The position of any wave at any time can be deduced directly from the diagram.
- The total potential at any point, at any instant of time is the superposition of all the waves which have arrived at that point up until that instant of time, displaced in position from each other by intervals equal to the difference in their time of arrival.
- The history of the wave is easily traced. It is possible to find where it came from and just what other waves went into its composition.
- Attenuation is included, so that the wave arriving at the far end of a line corresponds to the value entering multiplied by the attenuation factor of the line.

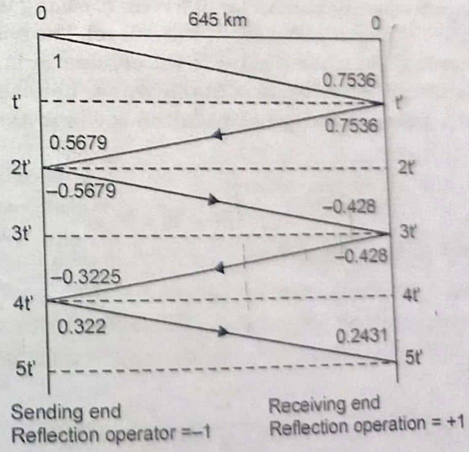


Fig. Lattice diagram for line with attenuation

Q. 2. (a) A transmission line has a span of 275 metres between level supports. The conductor has a diameter of 19.53 mm, weights 0.844kgf/m and has an

ultimate breaking strength of 7950 kgf. Each conductor has a radial has an
ultimate breaking strength of 7950 kgf. Each conductor has a radial covering of ice 9.53 mm thick and is subjected to a horizontal wind pressure of 40 kgf/ m^2 of the ice covered projected area. If the factor of safety is 2, calculate the deflected sag and the vertical component of the sag. One cubic metre of ice weighs 913.5 kgf. **(6.5)**

Ans. Refer Q.2. (a) First Term Examination 2018.

Q.2. (b) In brief write effects of different type of loadings on overhead conductors. **(6)**

Ans. Effect of wind and ice loading

In actual practice conductors are suspended to wind pressures and often loaded with ice coating in cold areas. The force due to wind is assumed to act horizontally to the conductor. And the force applied by ice loading is vertically downwards. Therefore, the total force on the conductor is the vector sum of vertical and horizontal forces.

In this case, weight of the conductor per unit length will be,

- $w_t = \text{sqrt}.[(w + w_i)^2 + (w_w)^2]$

where,

- w = weight of the conductor per unit length
- w_i = weight of the loaded ice per unit length
- w_w = wind force per unit length

When a conductor has wind as well as ice loading,

- The conductor sets itself in a plane at an angle θ to the vertical plane.

Where, $\tan\theta = w_w / (w + w_i)$

In this case, sag is given by $S = w_t.l^2 / 2T$. But, here, S represents the slant sag, i.e. sag is in the plane where the conductor has set itself. The vertical sag is equal to S.cos θ.

Q.3. (a) A transmission line conductor at a river crossing is supported from two towers at heights of 30m and 90m above water level. The horizontal distance between the towers is 270m. If the tension in the conductor is 1800 kgf and the conductor weighs 1.0 kgf per meter, find the clearance between the conductor and the water at a point midway between the towers. Assume Parabolic configuration. **(6.5)**

Ans. Difference in level between two support

$$h = 90 - 30 = 60 \text{ m}$$

Distance of lowest points of conductor

$$x = \frac{L}{2} - \frac{Th}{\omega L} = \frac{270}{2} - \frac{1800 \times 60}{1.0 \times 270} = -265$$

Distance of midpoint P from → $O = \frac{L}{2} - x = \frac{270}{2} - (-265)$

$$= 400 \text{ m}$$

Distance of B from → $O = L - x = 270 - (-265) = 535$ m

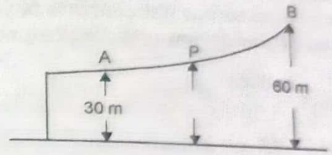
Height of midpoint P above O

$$S_{mid} = \frac{\omega\left(\frac{L}{2} - x\right)^2}{2T} = \frac{1.0 \times 400^2}{2 \times 1800} = 44.44 \text{ m}$$

Height of B from O

$$S_2 = \frac{\omega(L - x)^2}{2T} = \frac{1.0 \times 535^2}{2 \times 1800} = 79.51 \text{ m}$$

Hence mid point P is (79.51 - 44.44 = 35.07 m) below point B or (90 - 35.07 = 54.93 m) above water level.



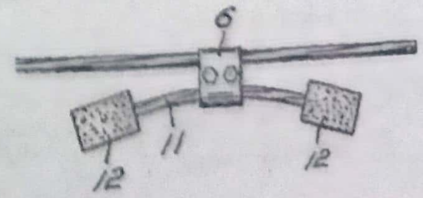
Q.3. (b) Write short notes on (Prevention of vibrations in conductors)
(i) Armour rods (ii) Stock bridge Damper **(6)**

Ans. (i) Armour rods: Armor rods are intended to protect against bending, compression, abrasion, arc over whilst also being capable of providing a repair function. The degree of protection needed on a specific line depends upon a number of factors such as line design, temperature, tension, exposure to wind flow and vibration history on a similar construction in the same area. Armor rods are recommended as a minimum protection for clamp type supports or suspension. Armor Rods may be used to restore full conductance and strength to AAC, AAAC and ACSR conductors, except high strength ACSR, where damage does not exceed 50% damage for 7 & 19 strand conductors or 25% damage for 37 & 61 strand conductors. Preformed Armor rods are extremely effective in relieving or suppressing conductor strains and therefore extending conductor service life.

Preformed Armor rods are chamfered and over a certain size are ball ended to create a smooth uniformed finish to minimise corona.

(ii) Stock bridge damper:

A **Stockbridge damper** is a tuned mass damper used to suppress wind-induced vibrations on slender structures such as overhead power lines and long cantilevered signs. The dumbbell-shaped device consists of two masses at the ends of a short length of cable or flexible rod, which is clamped at its middle to the main cable. The damper is designed to dissipate the energy of oscillations in the main cable to an acceptable level. Its distinctive shape gives it the nickname **"dog-bone damper"**



Vibrations in the main cable were passed down through the clamp and into the shorter damper, or "messenger", cable. This would flex and cause the symmetrically placed concrete blocks at its ends to oscillate. Careful choice of the mass of the blocks, and the stiffness and length of the damper cable would match the mechanical impedance of the damper to that of the line, and greatly attenuate oscillation of the main cable. Since Stockbridge dampers were economical, effective and easy to install, they became used routinely on overhead lines Live-line working using hot stick tools meant it was possible to retrofit dampers to lines while energised

Q. 4. A three phase, 132 kV, 50 Hz, 150 km long transmission line consists of three stranded aluminium conductors spaced triangularly at 3.8 m between centres. Each conductor has a diameter of 19.53 mm. the surrounding air is at a temperature of 30°C and at the barometric pressure of 750 mm of mercury. If the breakdown strength of air is 2.11 kV (rms) per cm and the surface factor is 0.85, determine the disruptive critical voltage. Also, determine the visual critical voltages for local and general corona if the surface factors are 0.72 and 0.82 for visual corona (local) and visual corona (general) respectively. (12.5)

Ans. Conductor radius = $\frac{19.53}{2} = 9.765\,\text{mm} = 0.9765\,\text{cm}$

Spacing of conductor d = 3.8 m = 380 cm

$$g_0 = 2.11 \text{ kV/cm rms}$$

$$\delta = \frac{3.92b}{273+t} = \frac{3.92\times 75}{273+30} = 0.9703$$

Critical disruptive voltage to neutral

$$V_{d0} = g_0\,\delta\, m_0 r \log_e \frac{d}{r}$$

$$= 2.11 \times 0.9703 \times 0.85 \times 0.9765 \log_e \frac{380}{0.9765}$$

$$= 4.40 \text{ kV (rms)}$$

Critical disruptive line to line voltage

$$= \sqrt{3}\times 4.40 = 7.62\,\text{kV(rms)}$$

Visual critical voltage to neutral

$$V_{v0} = g_0 \delta m_v r\left(1+\frac{0.3}{\sqrt{\delta r}}\right)\log_e \frac{d}{r}$$

For local corona

$$V_{v0} = 2.11\times 0.9703\times 0.72\times 0.9765\left(1+\frac{0.3}{\sqrt{0.9703\times 0.9765}}\right)$$

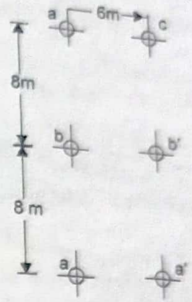
$$= 1.88 \text{ kV (rms)}$$

For general corona

$$V_{v0} = 2.11\times 0.9703\times 0.82\times 0.9765\left(1+\frac{0.3}{\sqrt{0.9703\times 0.9765}}\right)$$

$$= 2.14 \times \text{kV (rms)}$$

Q. 5. A 100 km double circuit transmission line with 7 strand copper conductors each strand having 5 mm diameter, has 6 conductors arranged in the form shown in fig. The line is transposed at regular intervals. Evaluate the GMD and GMR and calculate the overall inductance per phase of the line. (12.5)



Ans. Radius of each strand of conductor = $\frac{5}{2}$ mm = 2.5 mm

GMR of conductor r' = 2.176 × 2.5 mm = 5.44×10^{-3} m

Distance between conductors

$$d_{aa}' = \sqrt{6^2+16^2} = 17.089 \text{ m}: d_{bb} = 6\text{m}$$

$$d_{ac}' = d_{aa}'17.089 \text{ m} \;;\; d_{ab} = 8\text{m}$$

$$d_{ab}' = \sqrt{6^2+8^2} = 10\,\text{m}$$

GMR of phase a,

$$D_{ba} = \sqrt[4]{d_{ab} d_{a's}{}' d_{a'b} d_{aa'}}$$

$$= \sqrt{r' d_{set}} = \sqrt{5.44\times 10^{-3}\times 17.089} = 0.305\,m$$

Similarly,

$$D_{a'b} = \sqrt{t^4 \times d_{bb'}} = \sqrt{5.44\times 10^{-3}\times 6} = 0.1807\,\text{m}$$

and

$$D_{a'c} = \sqrt{r^4\times d_{cc}{}'} = \sqrt{5.44\times 10^{-3}\times 17.089} = 0.305\,\text{m}$$

Selt GMD of GMR.

$$D_a = \sqrt[4]{D_{ba} D_{ab} D_{bc}} = \sqrt[4]{0.305\times 0.1807\times 0.305}$$

$$= 0.2562 \text{ m}$$

$$D_{a'b} = \sqrt[4]{d_{a'b} D_{b'a} D_{a'b} D_{a'b'}} = \sqrt{d_{ab'} d_{ab'}} = \sqrt{8\times 10} = 8.944\,\text{m}$$

Similarly

$$D_b = \sqrt{d_{bc} = d_{bc'}} = \sqrt{8\times 10} = 8.944\,\text{m}$$

and

$$D_{bc} = \sqrt{d_{ca} d_{ca'}} = \sqrt{16\times 4} = 9.8\,\text{m}$$

$$D_{ca} \text{ or } D_{a,b} = \sqrt[4]{D_{a'b}D_{b'a}D_{a'b}} = \sqrt[3]{8.944\times 8.944\times 9.8} = 9.22m$$

Inductance per phase.

$$L = 0.2\log_e \frac{D_m}{Dl}\text{ mH/km} = 0.2\log\frac{9.22}{0.2562}$$

$$= 0.717 \text{ mH/km}$$

Inductance of 100 km line

$$= \frac{0.717\times 100}{1.000} = 0.0717\text{H}$$

Inducatance per phase km can also be determined form Eq. i.e.

$$L = 0.2\log_e 2^{1/a}\left(\frac{d_1}{r'}\right)^{1/2}\left(\frac{d_1^2+d_2^2}{4d_1^2+d_2^2}\right)^{1/6}\text{ mH}$$

or
$$L = 0.1\log_e 2^{1/3}\left(\frac{d_1}{r'}\right)\left(\frac{d_1^2+d_2^2}{4d_1^2+d_2^2}\right)^{1/3}\text{ mH}$$

where $d_1 = 8\text{m}, r = 5.44\times 10^{-3}\text{ m}; d_1^2+d_2^2 = 8^2+6^2 = 100\text{ m}^2$

and $4d_1^2+d_2^2 = 4\times 8^2+6^2 = 292\text{ m}^2$

So
$$L = 0.1,\ \log_e 2^{1/3}\left(\frac{8}{5.44\times 10^{-5}}\right)\left(\frac{100}{292}\right)^{1/3}$$

$$= 0.1\log_e 1.26\times 1.4706\times 0.6996 = 0.717\text{ mH}$$

Inductance of 100 km line

$$= 0.717\times 10^{-3}\times 100 = 0.0717\text{ H}$$

Q.6. (a) Explain Capacitive grading of cables. **(6.5)**

Ans. Capacitance grading: Capacitance grading is the process of using various layers of dielectrics with each dielectric having their own permittivity. The permittivity values should be in decreasing order from the surface of the conductor to the sheath of a cable. The product of permittivity of dielectric and radius from centre of the conductor to the particular layer of a dielectric should be constant at every layer of a dielectric.

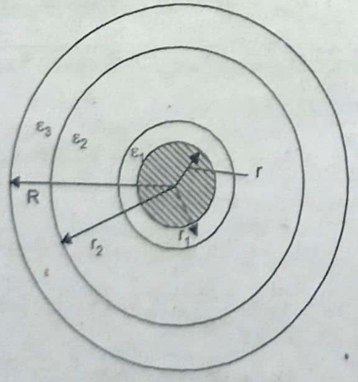


Fig.

Figure is the capacitance graded cable. Here, the radius of the conductor is r. Three dielectric layers are used in this cable. Consider the permittivity of first dielectric layer is ε_1, the distance from the surface of the conductor to first layer is r_1, the permittivity of second dielectric is ε_2, the distance from the surface of the conductor to second layer is r_2, the permittivity of third dielectric is ε_3, the distance from the surface of the conductor to third layer is R.

The relative permittivity values and their distances are $\varepsilon_1 > \varepsilon_2 > \varepsilon_3$ and $r_1 > r_2 > R$. The uniform dielectric stress can be achieved by maintaining the product of permittivity and radius of each dielectric as same, $\varepsilon_1 r_1 = \varepsilon_2 r_2 = \varepsilon_3 R$. The uniform dielectric stress cannot be achieved by capacitance grading alone. By capacitance grading alone an infinite number of dielectrics will be needed to achieve uniform dielectric stress. But it is not practically possible.

Inter sheath grading:

In this grading of a cable, a homogeneous dielectric is used. This homogenous dielectric is divided into several layers. Mechanical inter sheaths are placed in between sheaths and conductors. The inter sheaths are then held at adequate potentials which are placed in between conductor potential and earth potential. However, fixing of potentials at inter sheaths is a difficult task.

Q.6. (b) Determine the overall diameter of a single core cable and its most economical diameter when working on a three phase, 275 kV system. The maximum permissible stress in the dielectric is not exceed 15kV/mm. **(6)**

Ans. Peak value of cable voltage $V = 275\times\sqrt{2} = 388.91\,kV$

Max permissible stress $g_{max} = 15\text{ kV/mm}$

$= 150\text{ kV/cm}$

Most economical diameter $= \frac{2V}{g_{max}} = 5.19\text{cm}$

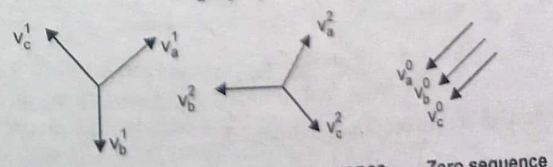
Q.7. (a) Prove that the three line voltage phasors V_{ab} V_{bc} and V_{ca} will have no zero sequence component and also prove that neutral current can flow only if zero-sequence currents exist. **(6.5)**

Ans.
$$V_a = V_a^0 + V_a^1 + V_a^2$$
$$V_b = V_b^0 + V_b^1 + V_b^2$$
$$V_c = V_c^0 + V_c^1 + V_c^2$$

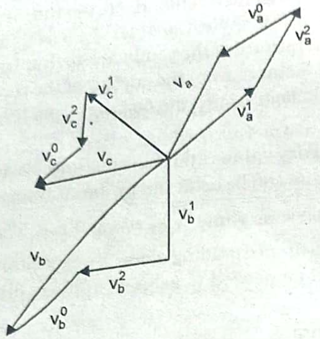
$V_a^0,\ V_b^0, V_c^0$ = zero sequence voltage

$V_a^1 V_b^1, V_c^1$ = Positive sequence voltage

V_a^2, V_b^2, V_c^2 = negative sequence voltage



Positive sequence components   Negative sequence components   Zero sequence components



$$V_b^0 = V_a^0 \qquad V_c^0 = V_a^0$$
$$V_b^1 = a^2V_a^1 \qquad V_c^1 = aV_a^1$$
$$V_b^2 = aV_a^2 \qquad V_c^2 = a^2V_a^2$$

$$\begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & a^2 & d \\ 1 & a & a^2 \end{bmatrix} \begin{bmatrix} V_a^0 \\ V_b^1 \\ V_c^1 \end{bmatrix} = 0 \begin{bmatrix} V_a^0 \\ V_b^1 \\ V_c^2 \end{bmatrix}$$

$$A^{-1} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix}$$

$$\begin{bmatrix} V_1^0 \\ V_2^1 \\ V_3^2 \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & 1 & 1 \\ 1 & a & a^2 \\ 1 & a^2 & a \end{bmatrix} \begin{bmatrix} V_a \\ V_b \\ V_c \end{bmatrix}$$

$$\Rightarrow \qquad V_a^0 = \frac{1}{3}(V_a + V_b + V_c) = 0$$

$\Rightarrow$ Line voltage phasors will have no zero sequence component

$$V_{ab} = V_a - V_b$$
$$V_{bc} = V_b - V_c$$
$$V_{ca} = V_c - V_a$$
$$V_{ab} + V_{bc} + V_{ca} = 0$$
$$V_{ab}^0 = \frac{V_{ab} + V_{bc} + V_{ca}}{3} = 0$$

(From definition of zero sequence component

$$I_n = I_a + I_b + I_c$$
$$= (I_a^0 + I_b^0 + I_c^0) + (I_a^1 + I_b^1 + I_c^1) + (I_a^2 + I_b^2 + I_c^2)$$

$$= I_a^0 + I_b^0 + I_c^0 = 3I_a^0$$
$$I_n = 3I_a^0$$

Since the positive sequence currents and negative sequence currents add separately to zero at neutral point n, there can not be any positive sequence or negative sequence currents in the connections from neutral to ground regardless of value of Z_n.

Q.7. (b) Mention all unsymmetrical faults that occur in power system. Explain any three of them. (6)

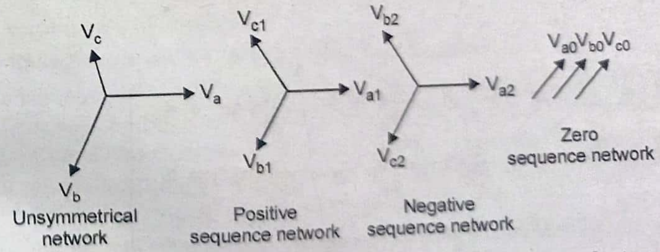
Ans. Unsymmetrical faults

Unsymmetrical faults involve only one or two phases. In unsymmetrical faults the three phase lines become unbalanced. Such types of faults occur between line-to-ground or between lines. An unsymmetrical series fault is between phases or between phase-to-ground, whereas unsymmetrical shunt fault is an unbalanced in the line impedances. Shunt fault in the three phase system can be classified as;

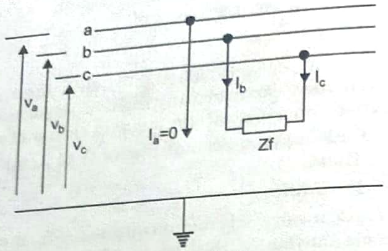
- Single line-to-ground fault (LG).
- Line-to-line fault (LL).
- Double Line-to-ground fault (LLG).
- Three-phase short circuit fault (LLL).
- Three-phase-to-ground fault (LLLG).

1. Single line to ground fault is the most frequently occurring fault (60 to 75% of occurrence). This fault will occur when any one line is in contact with the ground. Double line fault occurs when two lines are short circuited. This type of fault occurrence ranges from 5 to 15%. Double line to ground fault occurs when two lines are short circuited and is in contact with the ground. This type of fault occurrence ranges from 15 to 25% of occurrence.

- Symmetrical Components:
- Symmetrical components are derived to analyze unsymmetrical faults. The unsymmetrical network can be expressed in terms of three linear symmetrical components. The three symmetrical components are positive sequence component, negative sequence component and zero sequence component.
- Figure is the symmetrical components derived from an unsymmetrical or unbalanced network.



A line to line fault or unsymmetrical fault occurs when two conductors are short circuited. In the figure shown below shows a three phase system with a line-to-line fault phases b and c. The fault impedance is assumed to be Z_f. The LL fault is placed between lines b and c so that the fault be symmetrical with respect to the reference phase a which is un-faulted.



Line-to-Line Fault

Q.8. Write computational procedure for Newton Raphson method. (12.5)

Ans. Basic Newton-Raphson (NR) Techniques

Before discussing the application of NR technique in load flow solution. Let us first review the basic procedure of solving a set of non-linear algebraic equation by means of NR algorithm. Let there be '*n*' equations in '*n*' unknown variables $x_1, x_2, \ldots\ldots x_n$ as given below.

$$\left.\begin{aligned} f_1(x_1,x_2,\ldots\ldots x_n) &= b_1 \\ f_2(x_1,x_2,\ldots\ldots x_n) &= b_2 \\ \vdots \quad & \quad \vdots \\ \vdots \quad & \quad \vdots \\ f_n(x_1,x_2,\ldots\ldots x_n) &= b_n \end{aligned}\right\} \quad \ldots\text{(i)}$$

In equation (i), the quantities $b_1, b_2 \ldots b_n$ as well as the functions $f_1, f_2 \ldots f_n$ are known. To solve equation (i), first we take an initial guess of the solution and let these initial guesses be denoted as, $x_1^{(0)}, x_2^{(0)}, \ldots\ldots x_n^{(0)}$. Subsequently, first order Taylor's series expansion (neglecting the higher order terms) is carried out for these equation around the initial guess of solution. Also let, the vector of initial guess be denoted as $x^{(0)} = \left[x_1^{(0)}, x_2^{(0)}, \ldots\ldots x_n^{(0)}\right]^T$. Now, application of Taylor's expansion on the equations of set (i) yields.

$$\left.\begin{aligned} f_1\left(x_1^{(0)},x_2^{(0)},\ldots x_n^{(0)}\right)+\frac{\partial f_1}{\partial x_1}\Delta x_1+\frac{\partial f_1}{\partial x_2}\Delta x_2+\ldots+\frac{\partial f_1}{\partial x_n}\Delta x_n &= b_1 \\ f_2\left(x_1^{(0)},x_2^{(0)},\ldots x_n^{(0)},\right)+\frac{\partial f_2}{\partial x_1}\Delta x_1+\frac{\partial f_2}{\partial x_2}\Delta x_2\ldots+\frac{\partial f_2}{\partial x_n}\Delta x_n &= b_2 \\ \vdots \qquad\qquad & \quad \vdots \\ \vdots \qquad\qquad & \quad \vdots \\ f_n\left(x_1^{(0)},x_2^{(0)},\ldots x_n^{(0)},\right)+\frac{\partial f_n}{\partial x_1}\Delta x_1+\frac{\partial f_n}{\partial x_2}\Delta x_2+\ldots\ldots+\frac{\partial f_n}{\partial x_n}\Delta x_n &= b_n \end{aligned}\right\} \quad \ldots\text{(ii)}$$

Equation (ii) can-be written as,

$$\begin{bmatrix} f_1(X^{(0)}) \\ f_2(X^{(0)}) \\ \vdots \\ f_n(X^{(0)}) \end{bmatrix} + \begin{bmatrix} \frac{\partial f_1}{\partial x_1} & \frac{\partial f_1}{\partial x_2} & \cdots & \frac{\partial f_1}{\partial x_n} \\ \frac{\partial f_2}{\partial x_1} & \frac{\partial f_2}{\partial x_2} & \cdots & \frac{\partial f_2}{\partial x_n} \\ \vdots & \vdots & \vdots & \vdots \\ \frac{\partial f_n}{\partial x_1} & \frac{\partial f_n}{\partial x_2} & \cdots & \frac{\partial f_n}{\partial x_n} \end{bmatrix} \begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ \vdots \\ b_n \end{bmatrix} \quad \ldots\text{(iii)}$$

In equation (iii), the matrix containing the partial derivative terms is known as the Jacobin matrix (J). As can be seen, it is a square matrix. Hence, from equation (iii),

$$\begin{bmatrix} \Delta x_1 \\ \Delta x_2 \\ \vdots \\ \Delta x_n \end{bmatrix} = [\mathbf{J}]^{-1} \begin{bmatrix} b_1 - f_1(x^{(0)}) \\ b_2 - f_2(x^{(0)}) \\ \vdots \\ b_n - f_n(x^{(0)}) \end{bmatrix} = [\mathbf{J}]^{-1} \begin{bmatrix} \Delta m_1 \\ \Delta m_2 \\ \vdots \\ \Delta m_n \end{bmatrix} \quad \ldots\text{(iv)}$$

Equation (iv) is the basic equation for solving the 'n' algebraic equations given in equation (i). The steps of solution are as follow:

Step 1: Assume a vector of initial guess $X^{(0)}$ and set, iteration counter $k = 0$.

Step 2: Compute $f_1(X^{(k)}), f_2(X^{(k)}), \ldots\ldots f_n(X^{(k)})$.

Step 3: Compute $[\Delta m_1, \Delta m_2, \ldots\ldots |\Delta m_n|]$

Step 4: Compute error = max $[|\Delta m_1|, |\Delta m_2|, \ldots\ldots |\Delta m_n|]$

Step 5: If error $\leq \epsilon$ (pre. specified tolerance), then the final solution vector is $X^{(k)}$ and print the results. Otherwise go to step 6.

Step 6: Form the Jacobin matrix analytically and evaluate it at $X = X^{(k)}$.

Step 7: Calculate the correction vector $[\Delta x = [\Delta x_1, \Delta x_2, \ldots\ldots \Delta x_n]^T$ by using equation (iv).

Step 8: Update the solution vector $X^{(k+1)} = X^{(k)} + \Delta x$ and update $k = k + 1$ and go back to step 2.

Q. 9. Explain FDLF method. (12.5)

Ans. Fast-Decoupled load-flow (FDLF) technique

An important and useful of power system is that the change in real power is primarily governed by the charges in the voltage angles, but not in voltage magnitudes. On the other hand, the charges in the reactive power are primarily influenced by the charges in voltage magnitudes, but not in the voltage angles. To see this, let us note the following facts:

(a) Under normal steady state operation, the voltage magnitudes are all nearly equal to 1.0.

(b) As the transmission lines are mostly reactive, the conductances are quite small as compared to the susceptance ($G_{ij} << B_{ij}$).

(c) Under normal steady state operation the angular differences among the bus voltage are quite small ($\theta_i - \theta_j \approx 0$ (within $5° - 10°$)).

(d) The injected reactive power at any bus is always much less than the reactive power consumed by the elements connected to this bus when these elements are shorted to the ground ($Q_i << B_{ii}V_i^2$).

With these facts at hand, let us re-visit the equations for Jacobian elements in Newton Raphson (polar) method we have.

$$\frac{\partial P_i}{\partial V_j} = 2V_iG_{ii} + \sum_{\substack{k=1\\ \neq 1}}^{n} V_kY_{ik}\cos(\theta_i - \theta_k - \alpha_{ik})$$

$$= 2V_iG_{ii} + \sum_{\substack{k=1\\ \neq 1}}^{n} V_kY_{ik}[\cos(\theta_i - \theta_k)\cos\alpha_{ik} + \sin(\theta_i - \theta_k)\sin\alpha_{ik}]$$

$$= 2V_iG_{ii} + \sum_{\substack{k=1\\ \neq 1}}^{n} V_k[G_{ik}\cos(\theta_i - \theta_k) + B_{ik}\sin(\theta_i - \theta_k)]; j = i. \qquad \text{...(i)}$$

$$\frac{\partial P_i}{\partial V_j} = V_iY_{ij}\cos(\theta_i - \theta_j - \alpha_{ij})$$

$$= V_iY_{ij}[\cos(\theta_i - \theta_j)\cos\alpha_{ij} + \sin(\theta_i - \theta_j)\sin\alpha_{ij}]$$

$$= V_i[G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)]; j \neq i \qquad \text{...(ii)}$$

Now, G_{ii} and G_{ij} are quite small and negligible and also $\cos(\theta_i - \theta_j) \approx 1$ and $\sin(\theta_i - \theta_j) \approx 0$, as $[(\theta_i - \theta_j) \approx 0]$. Hence,

$$\frac{\partial P_i}{\partial V_i} \approx \text{ and } \frac{\partial P_i}{\partial V_j} \approx 0 \Rightarrow J_2 \approx 0 \qquad \text{...(iii)}$$

Similarly, from equation we get

$$\frac{\partial Q_i}{\partial \theta_j} = \sum_{\substack{k=1\\ \neq i}}^{n} V_iV_k[G_{ik}\cos(\theta_i - \theta_k) + B_{ik}\sin(\theta_i - \theta_k)]; j = i \qquad \text{...(iv)}$$

$$\frac{\partial Q_i}{\partial \theta_j} = -V_iV_j[G_{ij}\cos(\theta_i - \theta_j) + B_{ij}\sin(\theta_i - \theta_j)]; j \neq i \qquad \text{...(v)}$$

Again in light of the natures of the quantities G_{ii}, G_{ij} and $(\theta_i - \theta_j)$ as discussed above,

$$\frac{\partial Q_i}{\partial \theta_i} \approx 0 \text{ and } \frac{\partial Q_i}{\partial \theta_j} \approx 0 \Rightarrow J_3 \approx 0 \qquad \text{...(vi)}$$

Substituting equations (iii) and (iv) into equation (2.40) one can get,

$$\begin{bmatrix}\Delta P\\ \Delta Q\end{bmatrix} = \begin{bmatrix}J_1 & 0\\ 0 & J_4\end{bmatrix}\begin{bmatrix}\Delta\theta\\ \Delta V\end{bmatrix} \qquad \text{....(vii)}$$

In other words, ΔP depends only on $\Delta\theta$ and ΔQ depends only on ΔV. Thus, there is a decoupling between '$\Delta P - \Delta\theta$' and '$\Delta Q - \Delta V$' relations. Now, from equations we get

$$\frac{\partial P_i}{\partial \theta_j} = -\sum_{\substack{k=1\\ \neq i}}^{n} V_iV_kY_{ik}\sin(\theta_i - \theta_k - \alpha_{ik}); j = i$$

$$= V_iV_jY_{ii}\sin(\theta_i - \theta_i - \alpha_{ii}) - \sum_{k=1}^{n} V_iV_kY_{ik}\sin(\theta_i - \theta_k - \alpha_{ik}); j = i$$

$$= -B_{ii}V_i^2 - Q_i \approx -B_{ii}V_i^2;\ j = i[\text{as } Q_i << B_{ii}V_i^2] \qquad \text{....(viii)}$$

$$\frac{\partial P_i}{\partial \theta_j} = V_iV_jY_{ij}\sin(\theta_i - \theta_j - \alpha_{ij});\ j \neq i$$

$$= V_iV_jY_{ij}[\sin(\theta_i - \theta_j)\cos\alpha_{ij} - \cos(\theta_i - \theta_j)\sin\alpha_{ij}];\ j \neq i$$

$$= V_iV_j[G_{ij}\sin(\theta_i - \theta_j) - B_{ij}\cos(\theta_i - \theta_j);\ j \neq i$$

$$= -V_iV_jB_{ij};j \neq i \qquad \text{...(ix)}$$

Similarly from equations we get

$$\frac{\partial Q_i}{\partial V_i} = -2V_iB_{ii} + \sum_{\substack{k=1\\ \neq 1}}^{n} V_kY_{ik}\sin(\theta_i - \theta_k - \alpha_{ik}); j = i$$

or

$$\frac{\partial Q_i}{\partial V_j} = -2V_i^2B_{ii} + \sum_{\substack{k=1\\ \neq 1}}^{n} V_iV_kY_{ik}\sin(\theta_i - \theta_n - \alpha_{ik})j = i$$

or

$$\frac{\partial Q_i}{\partial V_j}V_i = -V_i^2B_{ii} + \sum_{\substack{k=1\\ \neq 1}}^{n} V_iV_kV_{ik}\sin(\theta_i - \theta_k - \alpha_{ik}) = Q_i - v_i^2B_{ii},\ j = i$$

or

$$\frac{\partial Q_i}{\partial V_j}V_i = -V_i^2B_{ii}; j = i\ [\text{as } Q_i << B_{ii}V_i^2]$$

or

$$\frac{\partial Q_i}{\partial V_i} = V_iB_{ii},\ j = i \qquad \text{...(x)}$$

$$\frac{\partial Q_i}{\partial V_j} = V_iY_{ij}\sin(\theta_i - \theta_j - \alpha_{ij});\ j \neq i$$

$$= V_iY_{ij}[\sin(\theta_i - \theta_j)\cos\alpha_{ij} - \cos(\theta_i - \theta_j)\sin\alpha_{ij}];\ j \neq i$$

$$= V_i[G_{ij}\sin(\theta_i - \theta_j) - B_{ij}\cos(\theta_i - \theta_j);\ j \neq i$$

$$\approx -V_iB_{ij}; j \neq i \qquad \text{...(xi)}$$

Combining equations (vii) (ix) we get $\Delta P_i = -V_i \sum_{k=1}^{n} V_k B_{ik} \Delta\theta_k$, Or,

$$\frac{\Delta P_i}{V_i} = -\sum_{k=1}^{n} V_k B_{ik} \Delta\theta_k, \quad ...(xii)$$

Now, as $V_i \approx 1.0$ under normal steady state operating condition equation (xii) reduces to,

$$\frac{\Delta P_i}{V_i} = -\sum_{k=1}^{n} B_{ik} \Delta\theta_k, \text{ Or, } \frac{\Delta P}{V} = [-B]\Delta\theta. \text{ Or,}$$

$$\boxed{\frac{\Delta P}{V} = [B']\Delta\theta} \quad ...(xiii)$$

Matrix B' is a constant matrix having a dimension of (n – 1) × (n – 1). Its elements are the negative of the imaginary part of the element (i, k) of the Y_{BUS} matrix where i = 2, 3, ...n and k = 2,3,....n.

Again combining equations (vii), (x) (ii) we get,

$$\Delta Q_i = -V_i \sum_{k=1}^{n} B_{ik} \Delta V_k \text{ Or, } \frac{\Delta Q_i}{V_i} = -\sum_{k=1}^{n} B_{ik} \Delta V_k \text{ Or,}$$

$$\boxed{\frac{\Delta Q}{V}[B'']\Delta V} \quad ...(xiv)$$

Again [B"] is also a constant matrix having a dimension of (n – m) × (n – m). Its elements are the negative of the imaginary part of the element (i, k) of the Y_{BUS} matrix where $i = (m + 1), (m + 2),....n$ and $k = (m + 1), (m + 2),....n$. As the matrixes [B'] and [B"] are constant, it is not necessary to invert these matrices in each iteration. Rather, the inverse of these matrices can be stored and used in every iteration, thereby making the algorithm faster. Futher simplification in the FDLF algorithm can be made by,

(a) Ignoring the series resistances is calculating the elements of [B']. Also by omitting the elements of [B'] that predominantly affect reactive power flows, i.e., shunt reactances and transformer off nominal in phase Laps.

(b) Omitting from [B"] the angle shifting effect of phase shifter, which prodominantly affects real power flow.

FIRST TERM EXAMINATION [FEB. 2019]
FOURTH SEMESTER [B.TECH]
POWER SYSTEM-I [ETEE–206]

Time : 1.5 hrs. **M.M. : 30**

Note :- *Attempt Q. No. 1 which is compulsory and any two more questions from remaining.*

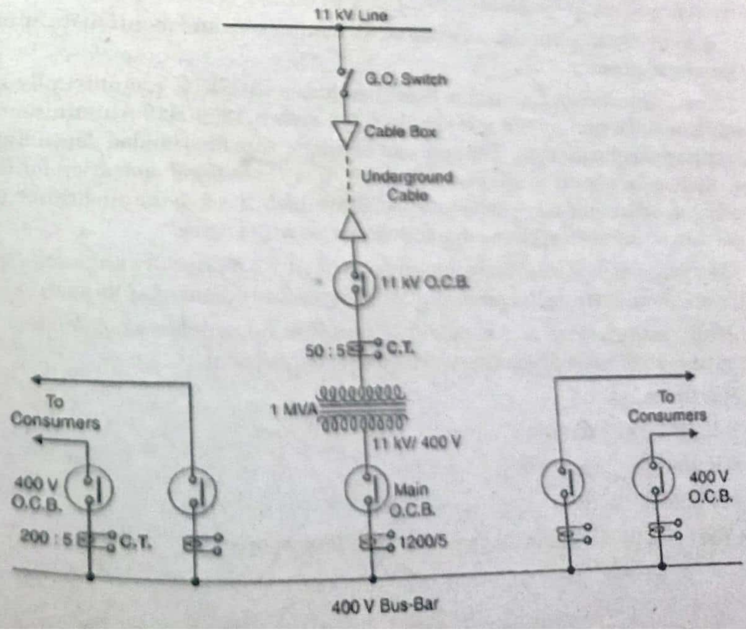
Q.1. (a) What are the advantages of per unit representation in power system? (2)

Ans. Per unit (pu) system advantages:

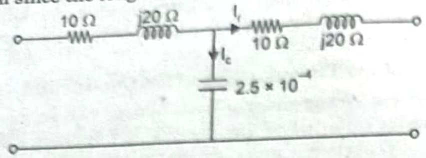
- Circuits are simplified
- Voltages have same range in per unit in all parts of the system from EHV system to distribution and utilization
- When expressed in the per unit system, apparatus parameters usually fall in narrow range regardless of apparatus size. For example, generator reactances in per unit are similar for both 100 MVA machines and 1000MVA machines. This facilitates data checking and hand calculations.
- For circuits connected by the transformers, per unit system is particularly suitable. By choosing suitable base kV for the circuits the per unit reactance remains the same, referred to either side of the transformer. Therefore the various circuits can be connected in the reactance diagram.
- This method is ideal for eliminate ideal transformers as circuit components since the typical power system contains hundreds, if not thousands of transformers and this is non trivial saving
- Sqrt (3) factor in the three phase circuit calculations is eliminated

Q.1. (b) Draw the single-line-diagram of an 11 KV substation? (2)

Ans.



These are per km since the length of Transmission line is 100 km



$\therefore$

$$I_r = \frac{20 \times 1000}{\sqrt{3} \times 110 \times 0.9} = 116.63 \text{ amp}$$

$$V_r = \frac{110 \times 1000}{\sqrt{3}} = 63508.52 \text{ Volts}$$

Taking I_r as reference

$$V_C = (63508.52 \times 0.9 + 116.63 \times 10) + j\,(63508.52 \times 0.435 + 116.63 \times 20)$$

$$= (57157.6 + 1166.3) + j\,(27626.20 + 2332.6)$$

$$= 58323.9 + j\,29958.80$$

$$I_C = j\omega CV_C = j\,314 \times 2.5\times10^{-4} \times (58323.9 + j\,29958.80)$$

$$= j\,4.578 - 2.351$$

$$I_s = 116.63 + j\,4.578 - 2.351$$

$$= 114.279 + j\,4.578 = 114.37 \angle 2.29$$

$$V_s = VC + I_s \frac{t}{2}$$

$$= (58323.9 + j\,29958.80) + (114.279 + j\,4.578) \times (10 + j\,20)$$

$$= (58323.9 + j\,29958.80) + (114.37 \angle 2.29) \times (22 \angle 63.43)$$

$$= (58323.9 + j\,29958.80) + 2508 \angle 66.72$$

$$= (58323.9 + j\,29958.80) + 991.22 + j\,2303$$

$$= 59315.12 + j\,32261.8$$

$$V_s = 67521 \angle 28.5$$

(ii) Efficiency $= \dfrac{\text{O/P Power}}{\text{O/P Power} + \text{losses}}$

$$= \frac{20 \times 10^6}{10^6 \times 20 + 3(116.63^2 \times 10 + 114.37^2 \times 10)}$$

$$= \frac{20 \times 10^6}{10^6 \times 20 + 3(136025.5 + 130804.969)}$$

$$= \frac{20 \times 10^6}{20 \times 10^6 + .8 \times 10^6}$$

$$= \frac{20}{20.8} = 96.15\%$$

END TERM EXAMINATION [MAY. 2019]
FOURTH SEMESTER [B.TECH]
POWER SYSTEM-I [ETEE-206]

Time : 3 hrs. **M.M. : 75**

Note :- *Attempt any five questions including Q.No. 1 which is compulsory. Assume missing data if any.*

Q.1. Attempt all the followings:-

Q.1. (a) How do the terms 'impedance drop', 'voltage drop' and 'voltage regulation', in connection with transmission lines differ? **(5)**

Ans. Impedance drop

Transmission line is a distributive network and lumped components takes place there. Such as lumped resistance, lumped inductance and lumped capacitance etc.

The nature of them is they do not follow Ohm's law and Kirchoff's law.

As the length increases then all lumped component increases and inductive reactance increases but capacitive reactance decreases.

Impedance are three types in transmission line..

1. Positive sequence impedance.
2. Negative sequence impedance .
3. Zero sequence impedance.

Positive sequence impedance resists the flow of positive sequence current at normal operating condition.

In electrical power systems, voltage regulation is a dimensionless quantity defined at the receiving end of a transmission line as:

$$\text{voltage regulation} = \frac{V_{nl} - V_{fl}}{V_{fl}}$$

where V_{nl} is voltage at no load and V_{fl} is voltage at full load. The percent voltage regulation of an ideal transmission line, as defined by a transmission line with zero resistance and reactance, would equal zero due to V_{nl} equaling V_{fl} as a result of there being no voltage drop along the line. This is why a smaller value of *Voltage Regulation* is usually beneficial, indicating that the line is closer to ideal.

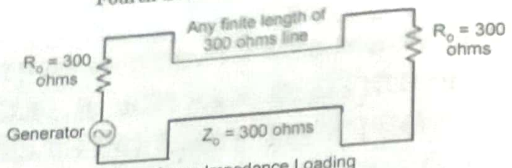
Q.1. (b) Why is it disadvantages to provide either too high sag or too low say? (3)

Ans. It Increases the Cost in Transmission Line when Too Much: The more space there is between the transmission towers, the more the transmission line will sag. If there is too much sag in a transmission line, it will increase the amount of conductor used, increasing the cost more than is necessary.

It Causes Power Failure: When a transmission line sag excessively, it is liable of causing power failure. An overheating electrical transmission line sagging into a tree sparked the greatest power failure in the Western United States in 1996. A similar incident is suspected to have caused the recent East Coast blackout. It means tight conductor and higher tension.

Q.1. (c) Explain surge impedance loading with respect to an overhead line? (3)

Ans. Surge impedance loading : The transmission line generates capacitive reactive volt-amperes in its shunt capacitance and absorbing reactive volt-amperes in its series inductance.The load at which the inductive and capacitive reactive volt-amperes are equal and opposite, such load is called surge impedance load.



Surge Impedance Loading

It is also called natural load of the transmission line because power is not dissipated in transmission. In surge impedance loading, the voltage and current are in the same phase at all the point of the line. When the surge impedance of the line has terminated the power delivered by it is called surge impedance loading.

Q.1. (d) List advantages of using per unit values in power system calculation? (3)

Ans. Refer to Q.1. (a) End Term Examination 2019.

Q.1. (e) The neutral grounding impedance Z_n appears as $3Z_n$ in the zero sequence equivalent circuit. Why? (3)

Ans. If V'_a, V'_b, and V'_c are assumed to be the phase potentials to earth at the neutral point, then

$$V'_a = V'_b = V'_c = V_n$$

Considering the a-phase alone,

$$V'_a = V_n = I_n Z_n = -3I_{a0}(Z_n) = -I_{a0}(3Z_n).$$

Also

$$V'_{a0} = -I_{a0} Z'_0.$$

Since neither positive- nor negative-phase sequence currents may flow in the neutral it follows that

$$V'_{a1} = -V'_{a2} = 0,$$

and so the only voltage appearing between N and earth must be of zero-phase sequence, i.e.

$$V_n = V'_a = V'_{a0} = -I_{a0}(3Z_n)$$

$$Z_0 = 3Z_n.$$

The zero-sequence impedance is thus three times the neutral impedance.

Q.1. (f) Why is one of the buses taken as slack bus in load flow studies? (3)

Ans. Slack Bus is defined in N-1 powerflow. Whenever your running powerflow, one bus is used as a slack or reference. Its a bus where all the mismatch can be absorbed after Gen = Load + Loss calculation during the iteration process. Its also known as a reference bus because all the final Bus Voltages and Bus angles are unknown initially. The reference or slack bus then is the only bus in poverflow where the bus voltage and angles variables are assumed to be 1 p.u and 0 radians angle when starting the iteration.

Q.1. (g) Why are ACSR conductors preferred for transmission and distribution lines? (5)

Ans. These conductors are formed by several wires of aluminum and galvanized steel, stranded in concentric layers. The wires which form the core are made by galvanized steel and external layer are of aluminum. It characteristics are, low weight combined within sigh breaking strength making the ma solution for long spans. Due to the great diameter of the conductor elect- tri closes of Corona Effect are reduced. ACSR cables are structurally plain. The lightness of aluminum contributes to effortless installation and the easy maintenance of the system.

The cost of setting up an aluminum conductor steel reinforced line is low. At the same time, the transmission capacity is large. These cables are characterized by excellent tensile strength (which means they can be utilized over a larger span without a lot of support being required), sufficient mechanical strength, applicability across voltage levels and suitability for both overhead transmission and distribution line utilization (versatility).

Q.2. (a) A string of four suspension insulators is connected across a 285 kV line. The self-capacitance of each unit is equal to 5 times pin to earth capacitance. Calculate: (6.5)

(i) the potential difference across each unit

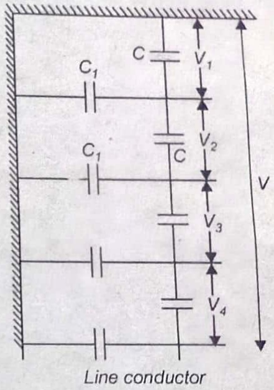
(ii) the string efficiency

Ans.

$$V = 285 \text{ kV}$$

$$K = \frac{C_1}{C} - \frac{1}{5}$$

$$C_1 = \frac{C}{5};$$



Let the voltage of conductor to earth be 'V' volts

$$V_2 = V_1(1+K) = V_1\left(1+\frac{1}{5}\right) = \frac{6}{5}V_1 = 1.2V_1$$

$$V_3 = V_1(1+3K+K^2) = V_1\left(1+\frac{3}{5}+\frac{1}{25}\right)$$

$$V_3 = \frac{51}{25}V_1$$

$$V_4 = V_1(1+6K+5K^2+K^3)$$

$$= V_1\left[1+\frac{6}{5}+\frac{1}{5}+\frac{1}{125}\right] = \frac{301}{125}V_1$$

$$\because \quad V = V_1+V_2+V_3+V_4$$

$$285 \text{ KV} = V_1+\frac{6}{5}V_1+\frac{51}{25}V_1+\frac{301}{125}V_1$$

$$\Rightarrow \quad V_1 = 42.87 \text{ kV};\ V_2 = 51.44 \text{ kV};\ V_3 = 87.45 \text{ kV};\ V_4 = 103.23 \text{ kV}$$

$$\text{String efficiency } \frac{V\times 100}{nV_n} = \frac{285}{4\times 103.23}\times 100 = 69.02\,\%$$

Q.2. (b) Show that in a string of suspension insulators, the disc nearest to the conductor has the hightest voltage across it? (6)

Ans. The voltage distribution across different units of an insulator string and string efficiency can be mathematically determined with the help of an equivalent circuit of the insulator string (Fig. 1) as below. Fig.1 shows the equivalent circuit of a string of suspension insulators containing 4 units.

Let the mutual capacitance between the links be C and shunt capacitance between links and earth be C_1, voltage across the first unit (nearest the cross-arm) be V_1, voltage across the second unit be V_2, voltage across the third unit be V_3, voltage across the fourth unit (nearest the line conductor) be V_4 and voltage between conductor and earth be V volts

Let $\frac{C_1}{C} = K$ or $C_1 = KC$

Applying Kirchhoff's first law to node A, we get

$$I_2 = I_1 + i_1$$

or $$\omega CV_2 = \omega CV_1 + \omega C_1 V_1$$

or $$\omega CV_2 = \omega CV_1 + \omega KCV_1$$

or $$V_2 = V_1(1 + K) \quad ...(1)$$

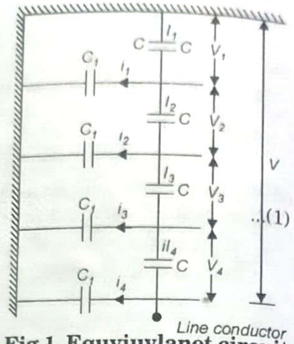


Fig.1. Equviuvlanet circuit for a 4-unit string

Applying Kirchhoff's first law to node B, we get

$$I_3 = I_2 + i_2$$

or $$\omega CV_3 = \omega CV_2 + \omega C_1(V_1 + V_2)$$

$\because$ voltage across the second shunt capacitance capacitance C_1 from the top

$$= V_1 + V_2$$

or $$\omega CV_3 = \omega CV_2 + \omega KC(V_1 + V_2)$$

or $$V_3 = V_2 + K(V_1 + V_2) = KV_1 + V_2(1 + K)$$

or $$V_3 = KV_1 + V_1(1 + K)(1 + K) \quad \because \text{From Eq. (1) } V_2 = V_1(1 + K) \quad ...(2)$$

or $$V_3 = V_1(1 + 3K + K^2)$$

Applying Kirchhoff's first law to node C, we get

$$I_4 = I_3 + i_3$$

or $$\omega CV_4 = \omega CV_3 + \omega C_1(V_1 + V_2 + V_3)$$

$\because$ Voltage across the third shunt capacitance C_1 from the top $= V_1 + V_2 + V_3$

or $$\omega CV_4 = \omega CV_1(1 + 3K + K^2) + \omega KC[V_1 + V_1(1 + K) + V_1(1 + 3K + K^2)]$$

or $$V_4 = V_1(1 + 6K + 5K^2 + K^3) \quad ...(3)$$

Finally voltage between line conductor and earth

$$V = V_1 + V_2 + V_3 + V_4$$

$$= V_1 + V_1(1 + K) + V_1(1 + 3K + K^2) + V_1(1 + 6K + 5K^2 + K^3)$$

$$= V_1(4 + 10K + 6K^2 + K^3) \quad ...(4)$$

Thus $$\frac{V_1}{1} = \frac{V_2}{1 + K} + \frac{V_3}{1 + 3K + K^2}$$

$$= \frac{V_4}{1 + 6K + 5K^2 + K^3}$$

$$= \frac{V}{4 + 10K + 6K^2 + K^3} \quad ...(5)$$

The greatest voltage will be obviously V_4 which is given as

$$V_4 = \frac{1 + 6K + 5K^2 + K^3}{4 + 10K + 6K^2 + K^3} V \quad ...(6)$$

and Percentage string efficiency

$$= \frac{V}{nV_4} \times 100 = \frac{V}{4 \times \frac{1 + 6K + 5K^2 + K^3}{4 + 10K + 6K^2 + K^3}} \times 100$$

Since $n = 4$, and flash-over voltage of one unit

= Greatest voltage across any unit i.e., V_4

or Percentage string efficiency

$$= \frac{4 + 10K + 6K^2 + K^3}{4(1 + 6K + 5K^2 + K^3)} \times 100 \quad ...(7)$$

Similarly derviation can be had for a string of insulators consisting of any number of units.

When the number of insulators in the string is large it becomes laborious to work out the voltage distribution across each unit, for such cases standard formula may be used.

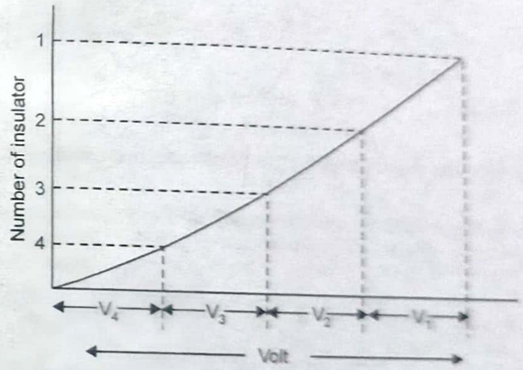


Fig. 2. Voltage Distribution Across Suspension Insulator String

Q.3. An overhead line has a span of 150 m between level supports. The conductor has a cross-sectional area of 2 cm². The ultimate strength is 5000 kg/cm2 and safety factor is 5. The specific gravity of the material is 8.9 gm/cc. The wind pressure is 1.5 kg/m. Calculate the height of the conductor above the ground level at which it should be supported if a minimum clearance of 7 m is to be left between the ground and the conductor? (12.5)

Ans. Span Length = 150 $wt = l$, Wind force/m; w_w = 1.5 kg/mt

Weight of conductor/m

$$w = \text{Conductor area} \times 100 \text{ cm} \times \text{Sp. gravity}$$

$$= 2 \times 100 \times 8.9 = 1780 \text{ gm} = 1.78 \text{ kg}$$

Working Tension; $T = 5000 \times \frac{2}{5} = 2000$ kg

Total weight of conductor/mt

$$w_t = \sqrt{w^2 + w_w^2} = \sqrt{(1.78)^2 + (1.5)^2} = 2.33 \text{ kg}$$

$$\text{Slant Sag} = \frac{w_t l^2}{8T} = \frac{2.33 \times (150)^2}{8 \times 2000} = 3.28 \text{ mt}$$

$$\text{Vertical Sag} = S\cos\theta = 3.28 \times w/w_t = 2.5 \text{ mt}$$

Conductor should be supported at height of = 7 + 2.5 = 9.5 mt

Q.4. (a) Find the disruptive critical and visual corona voltage of a grid line operating at 132kv. The following data is given: conductor diameter 1.9 cm, conductor spacing = 3.81 cm, temperature = 44°C, barometric pressure = 73.7 cm, conductor surface factor fine weather = 0.8, rough weather = 0.66? (6.5)

Ans. V = 132 KV $d = 3.81$ cm: $b = 73.7$ $r^{ot} = \frac{1.9}{2}$ cm $t = 44°C$

irregularity factor (fine weather) = 0.85; rough weather = 0.66

$$V_d = 21.1\, m/S_r \ln \frac{d}{r}$$

$$S = \frac{3.92b}{273+t} = \frac{3.92 \times 73.7}{273+44} = 0.99$$

$$V_d = 21.1 \times 0.85 \times .95 \times 0.99 \ln \frac{3.81}{.95}$$

$$V_d = 23.4 \text{ KV}$$

for rough weather $= .66 \times V_d$ for fine weather

$= .66 \times 23.4 = 15.4$ KV

Q.4. (b) Dervice the expression for the fault current for double line to ground fault? (6)

Ans. Fig.1. shows a Double Line to Ground Fault at F in a power system. The fault ay in general have an impedance Z^f as shown.

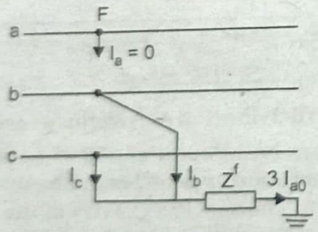


Fig. 1. Double line-to-ground (LLG) fault through impadance $-Z^f$

The current and voltage (to ground) conditions at the fault are expressed as

$$\left.\begin{array}{r} I_a = 0 \\ \text{or} \quad I_{a1} + I_{a2} + I_{a0} = 0 \end{array}\right\} \quad ...(1)$$

$$V_b = V_c = Z_f(I_b + I_c) = 3Zf I_{a0} \quad ...(2)$$

The symmetrical components of voltages are given by

$$\begin{bmatrix} V_{a1} \\ V_{a2} \\ V_{a0} \end{bmatrix} = \frac{1}{3}\begin{bmatrix} 1 & \alpha & \alpha^2 \\ 1 & \alpha^2 & \alpha \\ 1 & 1 & 1 \end{bmatrix}\begin{bmatrix} V_a \\ V_a \\ V_a \end{bmatrix} \quad ...(3)$$

from which it follows that

$$V_{a1} = V_{a2} = \frac{1}{3}[V_0 + (\alpha + \alpha^2)V_b] \quad (4a)$$

$$V_{a0} = \frac{1}{3}(V_a + 2V_b) \quad (4b)$$

From Eqs. (4a) and (4b)

$$V_{a0} - V_{a0} = \frac{1}{3}[2 - \alpha - \alpha^2]V_b = V_b = 3Z^f I_{a0}$$

$$V_{a0} = V_{a1} = 3Z^f I_{a0} \quad ...(5)$$

From Eqs. (1), (4a) and (5), we can draw the connection of sequence networks as shown in Figs. 2(a) and (b). The reader may verify this by writing mesh and nodal equations for these figures.

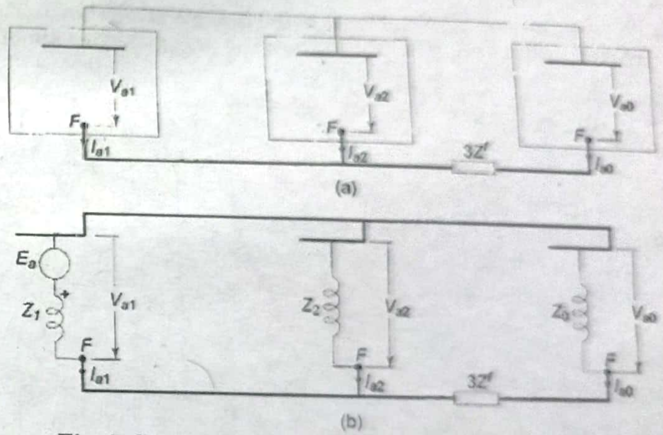


Fig. 2. Connection of sequence networks for a double line-to-ground (LLG) fault.

In terms of the Thevenin equivalents, we can write from Fig. 2(b) E_a

$$I_{a1} = \frac{E_a}{Z_1 + Z_2 \parallel (Z_0 + 3Z^f)}$$

$$= \frac{E_a}{Z_1 + Z_2(Z_0 + 3Z^f)/(Z_2 + Z_0 + 3Z^f)}$$

The above result can be obtained analytically as follows:

Substituting for V_{a1}, V_{a2} and V_{a0} in terms of E_a in Eq. and premultiplying both sides by Z^{-1} (inverse of sequence impedance matrix), we get

$$\begin{bmatrix} Z_1^{-1} & 0 & 0 \\ 0 & Z_2^{-1} & 0 \\ 0 & 0 & Z_0^{-1} \end{bmatrix}\begin{bmatrix} E_a - Z_1 I_{a1} \\ E_a - Z_1 I_{a1} \\ E_a - Z_1 I_{a1} + 3Z^f I_{a0} \end{bmatrix} = \begin{bmatrix} Z_1^{-1} & 0 & 0 \\ 0 & Z_2^{-1} & 0 \\ 0 & 0 & Z_0^{-1} \end{bmatrix}\begin{bmatrix} E_a \\ 0 \\ 0 \end{bmatrix}\begin{bmatrix} I_{a1} \\ I_{a2} \\ I_{a0} \end{bmatrix} \quad ...(6)$$

Premultiplying both sides by row matrix [1 1 1] and using Eqs. (1) and (2), we get

$$-\frac{3Z^f}{Z_0} I_{a0} + \left(1 + \frac{Z_1}{Z_0} + \frac{Z_1}{Z_2}\right) I_{a1} = \left(\frac{1}{Z_2} + \frac{1}{Z_0}\right) E_a \quad ...(7)$$

From Eq. 4(a), we have

$$E_a - Z_1 I_{a1} = -Z_2 I_{a2}$$

Substituting [See Eq. (1)]

$$I_{a2} = -(I_{a1} + I_{a0})$$

$$E_a - Z_1 I_{a1} = Z_2(I_{a1} + I_{a0})$$

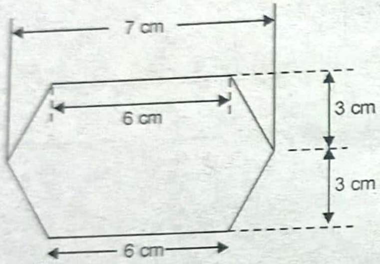
or

$$I_{a0} = \frac{E_a}{Z_2} - \left(\frac{Z_1 + Z_2}{Z_2}\right) I_{a1}$$

Substituting this value of I_{a0} in Eq. (7) and simplifying, we finally get

$$I_{a1} = \frac{E_a}{Z_1 + Z_2(Z_0 + 3Z^f)/(Z_2 + Z_0 + 3Z^f)}$$

Q.5. (a) A 3 phase double circuit line has the configuration shown. The radius of each conductor is 0.9 cm. Find the inductance per phase per km of the line length? (6.5)



Ans.

radius = 0.9 cm

$$D_{ab} = D_{bc} = \sqrt{3^2 + (0.5)^2} = 3.04 \text{ cm}$$

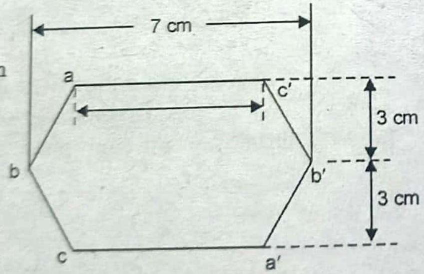
$$D_{ac} = 6 \text{ cm}$$

$$D_{ab'} = \sqrt{3^2 + (6.5)^2} = 7.15 \text{ cm}$$

$$D_{aa'} = \sqrt{6^2 + 6^2} = 8.48 \text{ cm}$$

$$D_{m'} = \sqrt[4]{D_{ab} D_{ac} D_{ab'} D_{ac'}}$$

$$= \sqrt[4]{3.04 \times 6 \times 7.15 \times 6}$$



$$D_{m1} = D_{m3} = 5.28 \text{ cm}$$

$$D_{m2} = \sqrt[4]{D_{ab} D_{bc} D_{ab'} D_{bc'}} = \sqrt[4]{3.04 \times 3.04 \times 7.15 \times 7.15}$$

$$= \sqrt{(3.04 \times 7.15)^2} = 4.66 \text{ cm}$$

$$D_m = \sqrt[3]{D_{m1} D_{m2} D_{m3}}$$

$$= \sqrt[3]{5.28 \times 5.28 \times 4.66}$$

$$= 5.06 \text{ cm}$$

$$= .0506 \text{ mt}$$

$$D_{s1} = \sqrt{0.9 \times 8.48} = D_{s3}$$

$$\boxed{D_{s1} = 2.76} \text{ cm}$$

$$D_{s2} = \sqrt{0.9 \times 7} = 2.5 \text{ cm}$$

$$D_s = \sqrt[3]{2.76 \times 2.76 \times 2.5} = 2.67 \text{ cm} = .0267$$

$$\text{Inductance} = 2 \times 10^{-7} \ln \frac{.0506}{.0267} = 1.278 \times 10^{-7} \text{ Henry/wt/phase}$$

Q.5. (b) Deduce the method of images for the calculation of capacitance of a 3 phase transmission line? (6)

Ans. For a system of charged conductors running parallel to the conducting plane, it may be assumed that each has got its own image located directly below it (Fig. 1). Each of them has a charge equal and opposite to the original conductor.

We shall utilize the method of images of illustrate the effect of earth on the capacitance of a two-wire line

Fig.2 shows a two-wire single-phase line having conductors a and b. The spacing between the conductors is D. a' and b' are images of a and b respectively. The charges on a and b are $+q$ and $-q$ respectively. The potential difference between a and b can be written as

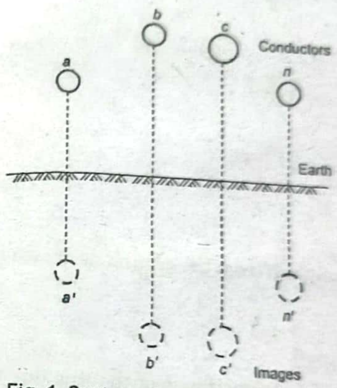


Fig. 1. System of conductors and their images.

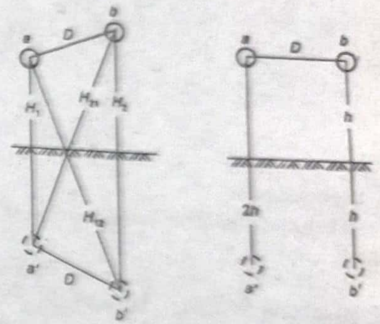


Fig. 2. Single-Phase Single-Circuit line.

$$V_{ab} = \frac{1}{2\pi\varepsilon_0}\left[q_a \ln\frac{D_{ab}}{D_{aa}} + q_b \ln\frac{D_{bb}}{D_{ba}} + q_a' \ln\frac{D_{a'b}}{D_{a'a}} + q_b' \ln\frac{D_{b'b}}{D_{b'a}}\right] \quad ...(1)$$

But,

$$q_a = q, q_b = -q_a = -q$$

$$q_{a'} = -q, q_b' = q$$

$$D_{aa} = D_{bb} = r$$

$$D_{ab} = D_{a'b'} = D, D_{aa'} = H_1, D_{bb'} = H_2$$

$$D_{ab'} = H_{12}, D_{ba'} = H_{21}$$

Substituting these values in Eq. (1) we get

$$V_{ab} = \frac{1}{2\pi\varepsilon_0}\left[q \ln\frac{D}{r} - q\ln\frac{r}{D} - q\ln\frac{H_{21}}{H_1} + q\ln\frac{H_2}{H_{12}}\right]$$

$$= \frac{1}{2\pi\varepsilon_0}\left(2q\ln\frac{D}{r} - q\ln\frac{H_{12}H_{21}}{H_1 H_2}\right)$$

$$= \frac{2q}{2\pi\varepsilon_0}\left(\ln\frac{D}{r} - \frac{1}{2}\ln\frac{H_{12}H_{21}}{H_1 H_2}\right)$$

$$= \frac{q}{\pi\varepsilon_0}\left(\ln\frac{D}{r} - \ln\frac{(H_{12}H_{21})^{1/2}}{(H_1 H_2)^{1/2}}\right)$$

We define the mean distances

$$H_s \triangleq (H_1 H_2)^{1/2},\ H_m \triangleq (H_{12} H_{21})^{1/2}$$

Then the expression for V_{ab} can be written as

$$V_{ab} = \frac{q}{\pi\varepsilon_0}\left(\ln\frac{D}{r} - \ln\frac{H_m}{H_s}\right) \quad ...(2)$$

The line-to-line capacitance

$$C_{ab} = \frac{q}{V_{ab}} = \frac{\pi\varepsilon_0}{\ln\frac{D}{r} - \ln\frac{H_m}{H_s}}\ F/m$$

Line-to-neutral capacitance

$$C_n = \frac{q}{V_{an}} = \frac{q}{\frac{1}{2}V_{ab}} = \frac{2\pi\varepsilon_0}{\ln\frac{D}{r} - \ln\frac{H_m}{H_s}}$$

$$= \frac{1}{18\times 10^9\left(\ln\frac{D}{r} - \ln\frac{H_m}{H_s}\right)}\ F/m \quad ...(3)$$

Special case

When the conductors a and b are at the same height h from the ground as shown in Fig. 2.

$H_1 = H_2 = 2h$, $\quad H_{12} = H_{21} = (D^2 + 4h^2)^{1/2}$

$H_s = (H_1/H_2)^{1/2} = 2h$, $\quad H_m = (H_{12}H_{21})^{1/2} = (D^2 + 4h^2)^{1/2}$

Equation 3 shows that there is a slight increase in capacitance of the line due to presence of earth. The effect diminishes as the height of the conductor above the earth is increased. It is not possible to calculate the capacitance accurately. Capacitance is affected by many variable factors such as the temperature, contour of the ground, vegetation in the vicinity of the line, presence of towers and other objects, etc. Because of these uncertainties, the effect of earth is negligible. The capacitance values obtained with an accuracy of about 0.5 to 1 percent are regarded satisfactory.

Q.6. Three generators are rated as following: (12.5)

Generator 1 : 100 MVA, 33 KV, 10% reactance

Generator 2 : 150 MVA, 32 KV, 8% reactance

Generator 3 : 110 MVA, 30 KV, 12% reactance

Choosing 200 MVA and 35 KV as base quantities. Compute per unit reactances of the 3 generators. Refer to these base quantities. Draw reactance diagram and mark per unit reactances. The 3 generators are connected to common bus bar.

Ans. Generator 1 : 100 MVA, 33 KV, 10% reactance = .1 p.u.

Generator 2 : 150 MVA, 32 KV, 8% reactance = .08 p.u.

Generator 3 : 110 MVA, 30 KV, 12% reactance = .12 p.u.

Base values; 200 MVA, 35 KV

$$Z_{2pn} = Z_{1pn}\left(\frac{S_{b2}}{S_{b1}}\right)\left(\frac{V_{b1}}{V_{b2}}\right)^2$$

$$\text{Generator 1; } X'' = .1\times\left(\frac{200}{100}\right)\times\left(\frac{33}{35}\right)^2 = .1777 \text{ p.u.}$$

$$\text{Generator 2; } X'' = .08\left(\frac{200}{150}\right)\times\left(\frac{32}{35}\right)^2 = .0891 \text{ p.u.}$$

$$\text{Generator 3; } X'' = .12\left(\frac{200}{110}\right)\times\left(\frac{30}{35}\right)^2 = .1602 \text{ p.u.}$$

$$V_{pu1} = \frac{33}{35} = .9428 \text{ p.u.}$$

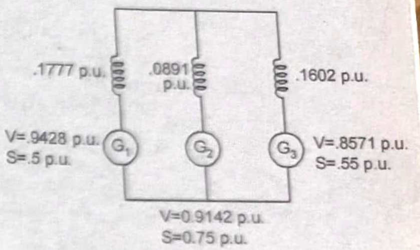
$$S_1 = \frac{100}{200} = .5 \text{ p.u.}$$

$$V_{pu2} = \frac{32}{35} = 0.9142 \text{ p.u.}$$

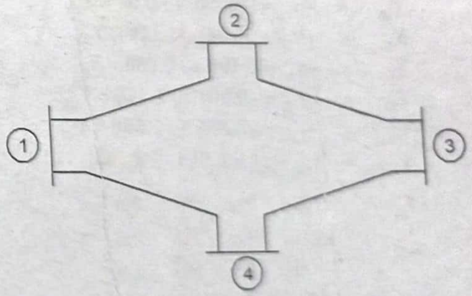
$$S_2 = \frac{150}{200} = 0.75 \text{ p.u.}$$

$$V_{pu3} = \frac{30}{35} = 0.8571 \text{ p.u.}$$

$$S_3 = \frac{110}{200} = 0.55$$



Q.7. (a) Find you bus for a 4- bus system? (6.5)

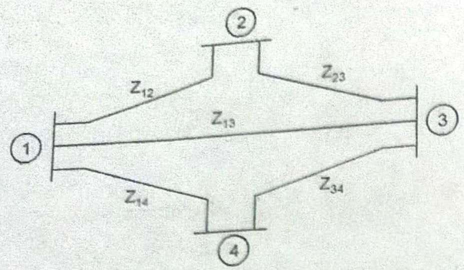


Line (bus to bus)	Impedance (p.u)
1-2	$0.25 + j\,1.0$
1-3	$0.20 + j\,1.8$
1-4	$0.30 + j\,1.2$
2-3	$0.20 + j\,0.8$
3-4	$0.15 + j\,0.6$

Ans.

$$y_{12} = \frac{1}{Z_{12}} = \frac{1}{0.25 + j1.0}$$

$$= 0.235 - j\,0.94$$



$$y_{13} = \frac{1}{Z_{13}} = \frac{1}{0.2 + j0.8}$$
$$= 0.294 - j1.176$$
$$y_{14} = \frac{1}{Z_{14}} = \frac{1}{0.3 + j1.2} = 0.196 - j\,0.784$$
$$y_{23} = \frac{1}{Z_{23}} = \frac{1}{0.2 + j0.8} = 0.294 - j\,1.176$$
$$y_{34} = \frac{1}{Z_{34}} = \frac{1}{0.15 + j0.6} = 0.392 - j\,1.568$$
$$Y_{11} = y_{12} + y_{13} + y_{14} = 0.725 - j\,2.9$$
$$Y_{22} = y_{12} + y_{23} = 0.529 - j\,2.116$$
$$Y_{33} = y_{13} + y_{23} + y_{34} = 0.98 + j\,3.92$$
$$Y_{12} = -y_{12} = -0.235 + j\,0.94 = Y_{21}$$
$$Y_{13} = -y_{13} = 0.294 + j1.176 = Y_{31}$$
$$Y_{14} = -y_{14} = 0.196 + j\,0.784 = Y_{41}$$
$$Y_{23} = -y_{23} = -0.294 + j\,1.176 = Y_{32}$$
$$Y_{34} = -y_{34} = -0.392 + j\,1.568 = Y_{43}$$
$$Y_{44} = y_{14} + y_{34} = 0.588 - j\,2.352$$

$$y_{bus} = \begin{bmatrix} y_{11} & y_{12} & y_{13} & y_{14} \\ y_{21} & y_{22} & y_{23} & y_{24} \\ y_{31} & y_{32} & y_{33} & y_{34} \\ y_{41} & y_{42} & y_{43} & y_{44} \end{bmatrix}$$

Q.7. (b) A 33 KV, 3 phase, 50 Hz underground cable, 3.4 km long uses 3 single core cables. Each cable has a core diameter of 2.5 cm and the radial thickness of insulation is 0.6 cm. The relative permittivity of the dielectric is 3.1. Find (i) maximum stress (ii) Total charging? (6)

Ans. System peak voltage per phase

$$V = \frac{33000}{\sqrt{3}} \times \sqrt{2} = 26{,}994 \text{ V}$$

Inner diameter of sheath

$$D = d + 2 \times \text{radial thickness of insulation}$$
$$= 2.5 + 2 \times 0.6 = 3.7 \text{ cm or .037 mt.}$$

Maximum electrostatic stress

$$g_{max} = \frac{2V}{d \log_e \frac{D}{d}} = \frac{2 \times 26{,}944}{0.025 \log_e \frac{0.037}{0.025}} = 5.5 \times 10^6 \text{ V/m}$$

Capacitance of cable

$$C = \frac{0.024\,E_s}{\log_{10} \frac{D}{d}} \times l \times 10^{-6} = \frac{0.024 \times 3.1}{\log_{10} \frac{0.037}{0.025}} \times 3.4 \times 10^{-6}$$
$$= 1.4857 \times 10^{-6} \text{ F}$$

Charging current of cable,

$$I_C = 2\pi f\, CVp = 2\pi \times 50 \times 1.4857 \times 10^{-6} \times 19052$$
$$= 8.893 \text{ Amp}$$

Total charging KVA $= \sqrt{3}\, V_L I_C \times 10^{-3}$

$$= \sqrt{3} \times 33 \times 10^3 \times 8.893 \times 10^{-3} = 508.23 \text{ KVAR}$$

Q.8. Write short notes on:

Q.8. (a) Bewley diagram (6.5)

Ans. Bewley's Lattice Diagram:

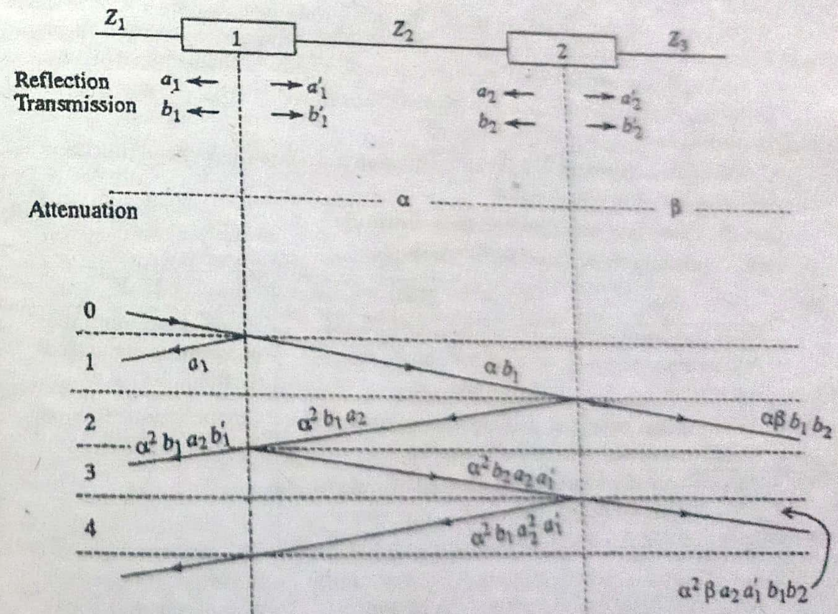


Fig. Reflection lattice of a travelling wave

In the arrangement shown in the figure, there are two junctions 1 and 2. Thetravel times for the waves are different through Z_1, Z_2, and Z_3. The lines with surge impedances Z_1, Z_2, and Z_3 are connected on either side of the junctions.

Let α and β be the attenuation coefficients for the two sections Z_2 and Z_3.

Let a_1 and a_1' be the reflection coefficients for the waves approaching from the left and the right at junction 1, and a_2 and a_2' be the corresponding reflection coefficients at junction 2.

Similarly, let b_1 and b_1' be the transmission coefficients for the waves that approach from the left and the right at junction 1, and the corresponding coefficients be b_2 and b_2' at junction 2. To construct the lattice diagram, the position 0 is taken when the wave coming from Z_1 reaches junction 1.

Junction 2 is taken to scale at the time interval equal to the travel time through the line Z_2 between the junctions 1 and 2.

In lattice diagram shown in Fig, horizontal intervals between junctions are laid as proportional to the time of passage of wave on these intervals and not proportional to their lengths. The advantage of doing this is that all diagonals, then, have the same slope and the time scale is applicable to every branch. The vertical time scale can suitably be chosen depending on the total time for which computations are required.

Then starting with the initial incident wave, various reflected and refracted waves are determined and indicated on the diagram till the lattice is completed. The characteristics of the lattice diagram are,

1. All waves travel downhill

2. The position of any wave at anytime is given by the time scale at the left of lattice.

3. The total potential at any place at any time is the super position of all the waves which have arrived at that point until the instant of time under consideration, displaced in position from each other by intervals equal to difference in their time of arrival.

4. The previous history of any wave can easily be traced. It is easy to find from where it came and what other wave constitute it.

5. The amount by which a wave is attenuated in travelling between junctions can also be incorporated in the diagram.

Q.8. (b) Comparison of power flow methods? (6)

Ans. Comparison of Load Flow Methods:

S.No.	G.S.	N.R.	FDLF
1.	Require large number of iterations to reach convergence.	Require less number of iterations to reach convergence.	Require more number of iterations than N.R. method
2.	Computation time per iteration is less.	Computation time per iteration is more.	Computation time per iteration is less.
3.	It has linear Convergence Characteristics.	It has quadratic conver gence Characteristics.	
4.	The number of iterations required for convergence increases with size of the system.	The number of iterations are independent of the size of the system.	The number of iterations iteration are does not dependent of the size of the system.
5.	Less memory requirements.	More memory requirements.	Less memory requirements than N.R. method.

Q.9. Explain the following:

Q.9. (a) Corona loss (6)

Ans. Refer Q.3.(a) End Term Examination 2019.

Q.9. (b) Type of insulator and their application? (6.5)

Ans. Types of Insulators in Transmission Lines should have the following desirable properties :

- High mechanical strength in order to withstand conductor load, wind load etc.
- High electrical resistance of insulator material in order to avoid leakage currents to earth.
- High relative permittivity of insulator material in order that dielectric strength is high.
- The insulator material be non-porous, free from impurities and cracks otherwise the permittivity will be lowered.
- **High ratio of puncture strength to flashover.**

The most commonly used material for insulators of overhead line is porcelain but glass, steatite and special composition materials are also used to a limited extent. Porcelain is produced by firing at a high temperature a mixture of kaolin, feldspar and quartz. It is stronger mechanically than glass, gives less trouble from leakage and is less effected by changes of temperature.

Types of Insulators:

The successful operation of an overhead Groo line depends to a considerable extent upon cond the proper selection of insulators. There are several Types of Insulators in Transmission Lines but the most commonly used are pin type, suspension type, strain insulator and shackle insulator.

1.Pin type insulators. The part section of a pin type insulator is shown in Fig. (i). As the name suggests, the pin type insulator is secured to the cross-arm on the pole. There is a groove on the upper end of the insulator for housing the conductor. The conductor passes through this groove and is bound by the annealed wire of the same material as the conductor.

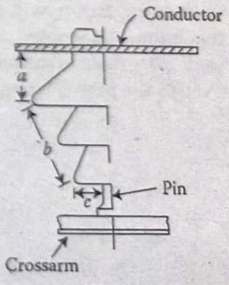


Fig.

Pin type insulators are used for transmission and distribution of electric power at voltages upto 33 kV. Beyond operating voltage of 33 kV, the pin type insulators become too bulky and hence uneconomical.

$$\text{Safety factor of insulator} = \frac{\text{Puncture strength}}{\text{Flash-over voltage}}$$

It is desirable that the value of safety factor is high so that flash-over takes place before the insulator gets punctured. For pin type insulators, the value of safety factor is about 10.

2. Suspension type insulators. The cost of pin type insulator increases rapidly as the working voltage is increased. Therefore, this type of insulator is not economical beyond 33 kV. For high voltages (>33 kV), it is a usual practice to use suspension type insulators shown in Fig. . They consist of a number of porcelain discs connected in series by metal links in the form of a string. The conductor is suspended at the bottom end of this string while the other end of the string is secured to the cross arm of the tower. Each unit or disc is designed for low voltage, say 11 kV. The number of discs in series would obviously depend upon the working voltage. For instance, if the working voltage is 66 kV, then six discs in series will be provided on the string.

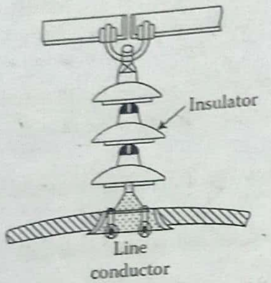


Fig. Suspension Insulator

3. Strain insulators. When there is a dead end of the line or there is corner or sharp curve, the line is subjected to greater tension. In order to relieve the line of excessive tension, strain insulators me used For low voltage lines (< 11 kV), shackle insulators are used as strain insulators. However, for high voltage transmission lines, strain insulator consists of an assembly of suspension insulators as shown in Fig. 8.8. The discs of strain insulators are used in the vertical plane. When the tension in lines is exceedingly high, as at long river spans, two or more strings are used in parallel.

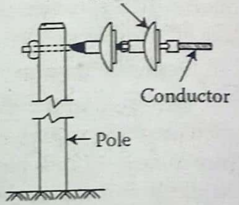


Fig. Strain insulator

4. Shackle insulators: In early days, the shackle insulators were used as strain insulators. But now a days, they are used for low voltage distribution lines. Such insulators can be used either in a horizontal position or in a vertical position. They can be directly fixed to the pole with a bolt or to the cross arm. Fig. shows a shackle insulator fixed to the pole. The conductor in the groove is fixed with a soft binding wire.

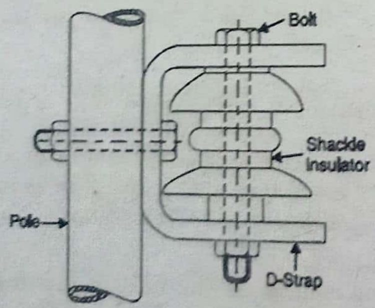


Fig. Shackle Insulator